

A project report on

**Integrating AI in Pharma: Revolutionizing Drug Discovery
and Formulation for Future Healthcare**

submitted to the Institute of Pharmacy, Bundelkhand University, Jhansi

in partial fulfillment of the award of the degree of

Bachelor of Pharmacy



2023-2024

UNDER THE SUPERVISION

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Institute of Pharmacy

2023-2024



Certificate

This is to certify that the project entitled “Integrating AI in Pharma: Revolutionizing Drug Discovery and Formulation for Future Healthcare” submitted in partial fulfillment of the requirement for the Bachelor of Pharmacy, Institute of Pharmacy, Bundelkhand University, Jhansi is a bonafide work carried out by **Sohan Lal** during the academic session 2023-2024.

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Declaration

This is to certify that the project entitled “**Integrating AI in Pharma: Revolutionizing Drug Discovery and Formulation for Future Healthcare**” is prepared by me under the direct guidance and supervision of **Dr. Sunil Kumar Prajapati** (Professor) Institute of Pharmacy, Bundelkhand University, Jhansi.

The same is submitted to Bundelkhand University, Jhansi in partial fulfillment of the requirement for the award of the degree of Bachelor of Pharmacy.

I further declare that I have not submitted this project report previously for the award of any degree or diploma to me.

Date: 22/05/2024

Sohan Lal

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1. Introduction

The process of drug development is complex and time-consuming, involving the identification of new therapeutic candidates that can effectively treat a range of ailments. The utilization of artificial intelligence has resulted in a substantial transformation in the methodology of drug discovery. AI has revolutionized the pharmaceutical sector by accelerating the drug development process, enhancing accuracy, and reducing expenses. This review will examine various AI techniques employed in drug discovery, such as machine learning (ML) for predicting drug properties, identifying potential drug candidates, and optimizing chemical structures. Deep learning (DL) is used to analyze extensive biological data, predict drug properties, and identify potential drug candidates. Natural language processing (NLP) is utilized to analyze scientific literature for potential drug candidates and generate drug summaries. Genetic algorithms (GA) are employed to generate new molecules that have the potential to be drug candidates. Network-based approaches are used to identify potential targets for drug development.

1.1. Historical Background of Artificial Intelligence in Medicine

AI in medicine refers to the application of artificial intelligence techniques, algorithms, and technologies in the field of medicine and healthcare. It entails utilizing computer systems and specialized software to scrutinize medical data, render choices, and execute duties typically carried out by human healthcare experts. The objective of AI in medicine is to enhance the precision, productivity, and efficacy of medical diagnosis, therapy, and patient care through the utilization of machine learning (ML), natural language processing (NLP), and other AI techniques.

Artificial intelligence (AI) is utilized in various aspects of medicine, including medical image processing, drug development, individualized treatment planning, disease diagnosis and prediction, virtual health assistants, electronic health record administration, and patient monitoring. AI systems can enhance healthcare workers' decision-making by analyzing extensive patient data and medical literature. They can identify trends and forecast patient outcomes, resulting in improved patient care and medical results. The domain of artificial intelligence in the realm of medicine is progressing swiftly and exerting influence in various domains like drug discovery, virtual health aides, and remote patient monitoring. AI-powered tools are anticipated to enhance medical diagnosis, disease prevention, and treatment outcomes, ultimately resulting in a more patient-centric and efficient healthcare system. The origins of AI in medicine may be traced back several decades, during which notable advancements in both AI and medical sciences have played a crucial role in its progression. Significant landmarks and occurrences in the historical progression of AI in medicine encompass the inception of the AI field during the 1950s by notable academics such as Alan Turing, who introduced the concept of intelligent machines [1]. Initial research in the field of artificial intelligence focused on replicating human problem-solving capabilities using formal logic and rule-based systems [1]. During the 1960s, the idea of "expert systems" was introduced. These systems involved transferring the knowledge and skills of human experts into computer programs to assist in decision-making within particular fields [1]. This established the groundwork for the implementation of artificial intelligence in the field of medicine. Early expert systems, including Dendral and MYCIN, significantly progressed the application of AI in the field of medicine. In the 1970s, there was a rise in the use of artificial intelligence (AI) in the field of medicine, specifically in the development of computer-aided diagnostic (CAD) systems for medical imaging AI [1]. MYCIN, an expert system designed for the purpose of diagnosing bacterial infections, showcased the capabilities of artificial intelligence in the field of healthcare [1]. During the 1980s, artificial intelligence (AI) image recognition algorithms were first used in medical imaging to assist in

analyzing X-rays, CT scans, and MRIs [1]. Artificial intelligence techniques, such as pattern recognition and machine learning, were included into medical imaging systems [1]. In the 1990s, there was advancement in natural language processing (NLP) which allowed artificial intelligence (AI) systems to comprehend and analyze medical text data [1]. The da Vinci Surgical System is an example of a robotic surgery system that integrates artificial intelligence and robotics to perform minimally invasive treatments [1]. As electronic health records (EHRs) became more common and the use of big data in healthcare increased, artificial intelligence (AI) applications also grew to manage large quantities of patient data [1]. Artificial intelligence (AI) in the field of medicine has started to play a role in customized medicine by using individual traits to forecast how patients will respond to treatments [1]. Deep learning (DL), which is a subset of machine learning (ML), has brought about a significant transformation in the field of artificial intelligence (AI) by revolutionizing many applications such as medical image analysis and illness diagnosis [1]. Artificial intelligence (AI) has been instrumental in the progress of precision medicine by customizing treatments according to genetic and molecular information [1]. These advancements underscore the substantial advancements achieved in the use of artificial intelligence in the field of medicine, which have paved the way for enhanced healthcare results and customized treatment methodologies.

1.2. Significance Statement

The historical context of AI in medicine demonstrates a gradual evolution from initial investigations to the present era of advanced AI applications in diverse areas of healthcare. AI technologies have the ability to transform the future of medicine and revolutionize healthcare procedures, benefiting both patients and healthcare professionals. In general, there has been a significant and continuous growth in the quantity of published papers on artificial intelligence (AI) over the years. The highest point was reached in 2023, with a total of 348,684 papers being published.

The quantity of published papers on AI-facilitated material discovery in medication delivery has experienced a notable rise in recent years, reaching a pinnacle of 1234 papers released in 2022. The quantity of published papers on artificial intelligence (AI) in the field of medicine has been steadily rising in recent years, reaching a pinnacle of 188,845 publications in 2021. The surge in the quantity of published papers on AI indicates a burgeoning interest in this technology and a corresponding increase in research activity in this field. The quantity of published papers on AI in medicine has also been rising in recent years, but at a more gradual rate (Figure 1). The rise in the quantity of published publications on AI-facilitated material discovery in medication delivery indicates a concurrent surge in the interest around the possible application of AI in this domain. This implies that there is still a considerable distance to cover before AI becomes broadly integrated into this particular domain.

Although there is limited coverage in the literature about the specific emphasis on AI-assisted material discovery in drug delivery, the research emphasizes the potential of AI in enhancing pharmaceutical product development, such as formulation design and drug discovery [2,3]. These observations indicate that AI has the potential to be a valuable instrument in speeding up the process of finding and improving materials for medication delivery. However, to obtain a more comprehensive analysis that specifically concentrates on AI-assisted material discovery in medication delivery, it would be advantageous to investigate further research articles or publications that directly tackle this subject matter.

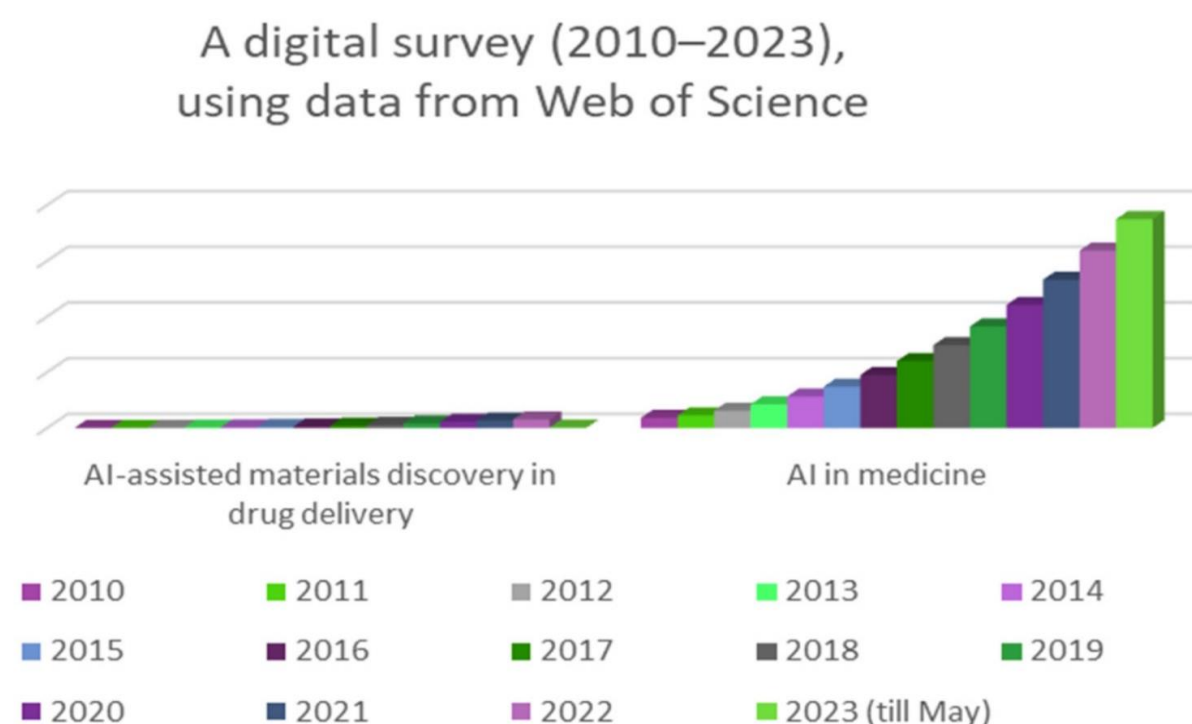


Figure 1. A comprehensive digital poll conducted from 2010 to 2023 provides compelling evidence to support the significance of the issue.

2. Role of AI in New Drug Discovery

Within the realm of medicine, there exist two distinct categories of AI applications: physical and virtual. Physical applications encompass robot-assisted surgery, AI-enhanced prosthetics, real-time patient monitoring, and automated laboratory processes.

For instance, artificial intelligence (AI) utilized in robot-assisted surgery can furnish medical practitioners with pertinent data to aid them in generating better-informed judgments. Although AI is incapable of substituting human doctors, it can augment their talents and enhance the quality of patient care. AI-powered surgical robots enhance surgeons' ability to execute intricate procedures with enhanced precision, control, and adaptability. These robots have the ability to decrease the likelihood of problems, decrease the level of invasiveness, and shorten the amount of time needed for recovery, resulting in improved surgical results [4]. Conversely, prosthetics powered by artificial intelligence are specifically created to adjust to the user's motions and react to their cerebral impulses. Advanced prosthetics greatly enhance the quality of life for individuals who have had amputation, enabling them to effortlessly and naturally carry out intricate tasks. AI-powered surveillance systems continuously evaluate patient data, including vital signs and electronic health records, to detect possible indications of worsening or complications. This allows healthcare providers to promptly intervene and prevent negative occurrences. Research has demonstrated that AI systems can surpass human doctors in specific diagnostic tasks, such as identifying particular forms of cancer or interpreting pulmonary function tests [5]. Automated laboratory operations, such as AI-powered robotic systems, efficiently streamline and automate many tasks in the laboratory, including sample examination, sorting, and preparation. This decreases the burden on laboratory personnel and lessens the likelihood of human mistakes, hence guaranteeing more precise outcomes.

Drug Discovery and Development are classified under the Virtual Applications category, along with diagnostic help, individualized treatment regimens, and virtual health assistants. Virtual artificial intelligence (AI) apps assist healthcare practitioners in diagnosing diseases with more accuracy and efficiency. Artificial intelligence algorithms have the capability to examine medical imaging data, including X-rays, CT scans, and MRI pictures, in order to identify irregularities and aid in the early diagnosis of diseases [6]. This capacity results in a substantially decreased likelihood of misdiagnosis and contributes to improved patient outcomes. The strength of AI resides in its capacity to efficiently collect and analyze vast quantities of medical data, identifying patterns that may not be readily apparent to human observers. By doing this, artificial intelligence (AI) can enhance diagnostic precision and provide individualized treatment strategies. Deep learning algorithms have the capability to detect abnormalities or possible illnesses in medical images, hence aiding radiologists in their analysis [6]. AI has demonstrated its utility in the field of gastroenterology by accurately identifying aberrant structures in endoscopy and ultrasound pictures, such as colonic polyps [7]. In addition, wearable gadgets driven by artificial intelligence can remotely monitor patients and provide healthcare professionals with real-time data, thereby presenting potential for early intervention. These gadgets have been created to detect and inform caretakers about seizures in people with epilepsy [8]. AI-driven virtual applications have the capability to assess an individual's health data, including their genetics [9], medical history [10], lifestyle factors [11], and present health condition, in order to generate customized treatment recommendations. These plans can enhance the efficiency of drug dose, anticipate the patient's response to therapy, and suggest specific remedies, so assuring more effective and tailored care for the patient. AI can enhance precision medicine through the analysis of a patient's genetic data and medical history, enabling the prediction of illness risk, the identification of best treatment strategies, and the discovery of possible drug targets. In addition, AI-powered virtual health assistants provide round-the-clock support and individualized health advice to patients [12]. They have the ability to respond to medical inquiries, send reminders to patients regarding their drug schedules, and offer suggestions for lifestyle choices, thereby encouraging patient involvement and proactive management of healthcare.

AI algorithms aid researchers in the **drug discovery and development** process by examining extensive databases of molecular structures, biological interactions, and clinical trial data to uncover possible drug candidates. This expedites the drug discovery process and shows potential for an expedited development of novel drugs.

AI in drug discovery refers to the application of artificial intelligence techniques and algorithms to the process of discovering new drugs.

(i) ***Identification and validation of targets***

Artificial intelligence has had a profound influence on the field of drug discovery, namely in the domains of target identification and validation. This process entails the identification of potential biological targets and the clarification of their roles in diseases. Subsequently, these targets are validated to ensure their direct involvement in disease mechanisms and the likelihood of therapeutic effects through target modulation. This process plays a critical role in identifying potential drug targets by analyzing genomic, proteomic, and metabolomic data (Figure 2). Machine learning algorithms analyze extensive datasets to identify the proteins or biological processes associated with particular diseases, providing researchers with vital information for the development of drugs [13].

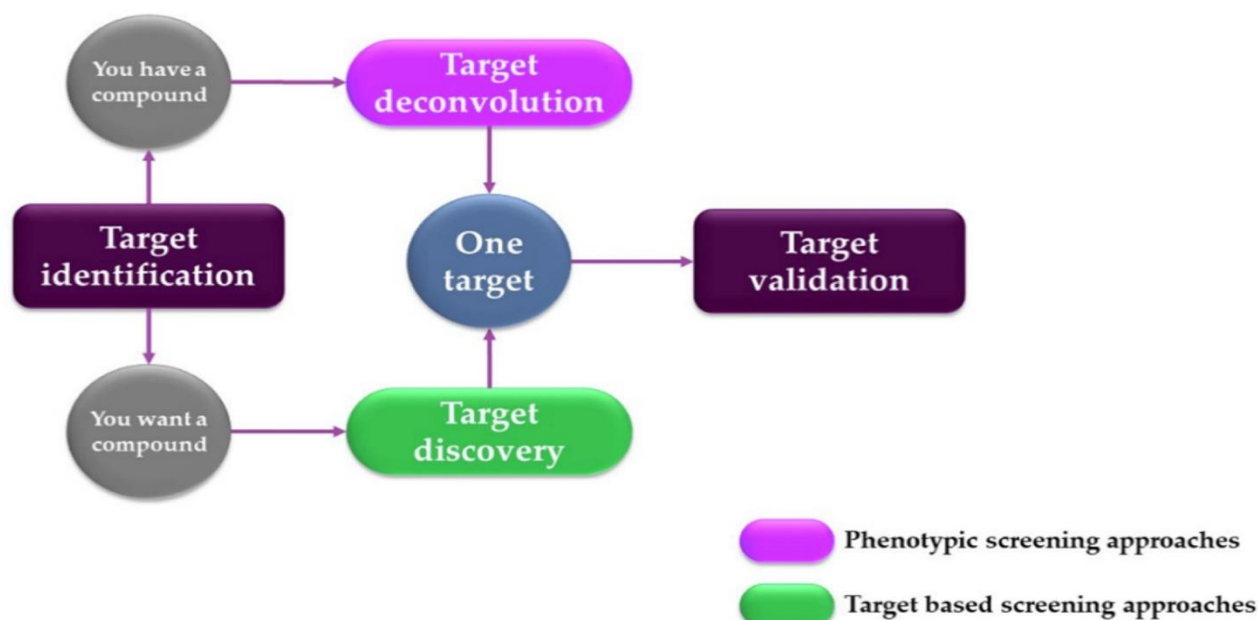


Figure 2. The drug development process involves two main approaches: target deconvolution and target discovery.

For example, ML-based methods like Kronecker regularized least squares (KronRLS) assess the resemblances between medications and protein molecules in order to predict drug-target binding affinity (DTBA). SimBoost use regression trees to forecast DTBA, taking into account both feature-based and similarity-based interactions [14].

AI also assists in the identification of the target. An ideal target should possess druggable characteristics, ensuring its suitability for pharmaceutical intervention. Additionally, it should have a high level of safety, efficiency, and the ability to meet commercial demands. Nevertheless, novel approaches for treating diseases involve areas that were previously more challenging to address. Target validation is an essential stage in the process of drug discovery since it confirms that a certain molecular target is directly implicated in the mechanism of a disease and that manipulating the target is likely to provide a therapeutic effect [15]. Target validation encompasses various methods, including analyzing the structure-activity relationship, manipulating target genes through knockdown or overexpression, creating drug-resistant mutants of the target, utilizing degradation-based tools to predict target effects, and monitoring downstream signaling pathways [16]. Artificial intelligence (AI) has been utilized to forecast drug-target interactions [17], quantify the binding affinity of a medication [18], and identify and authenticate targets [19].

(ii) **Virtual screening and drug design**

AI-driven virtual screening tools utilize advanced algorithms to assess the three-dimensional structures of target proteins and accurately forecast the interactions between prospective therapeutic compounds and these proteins. This accelerates the drug design process and enables researchers to select potential therapeutic candidates for subsequent testing [20].

Virtual drug screening is a computational method that employs artificial intelligence to forecast the effectiveness of new medications by matching chemical structures to targets (Figure 3). This technique enables scientists to quickly assess a collection of substances for their ability to attach to and hinder certain receptor or enzyme targets [21]. Artificial intelligence algorithms have the capability to examine molecular structures, forecast binding affinities, and determine the priority of

compounds for additional experimental testing [21]. In addition, molecular docking simulations can employ AI approaches such as Bayesian docking approximations and RL [22].

Molecular docking is the process of anticipating the optimal arrangement of a small molecule (drug candidate) when it is attached to a target protein. AI algorithms have the ability to analyze the conformational space and make predictions about the binding affinity between a medication and its target protein [22]. Machine learning algorithms for drug design can be trained using extensive datasets containing known drug-target interactions. This training allows the algorithms to predict new drug-target pairs or enhance the characteristics of existing medications. These algorithms have the ability to acquire knowledge about the patterns and connections between chemical structures and biological activities, which allows for the identification of new potential drugs [23]. Graph neural networks (GNNs) have been employed to forecast molecular characteristics and enhance medication design [24]. Graph neural networks (GNNs) are being used more and more in drug discovery due to their capacity to precisely forecast chemical characteristics. These sophisticated AI models have exceptional proficiency in comprehending the intricate arrangements of molecules, facilitating a more effective and precise approach to medication creation. Graph Neural Networks (GNNs) assess and anticipate the behavior of molecular structures by modeling them as graphs, with atoms as nodes and chemical bonds as edges. This enables the optimization of drug design procedures. Artificial intelligence can optimize high-throughput virtual screening, a process that includes analyzing extensive databases of substances to uncover promising drug candidates. Machine learning algorithms have the ability to examine chemical characteristics, structure-activity connections, and other molecular attributes in order to give priority to compounds that have a high likelihood of being further investigated [25].

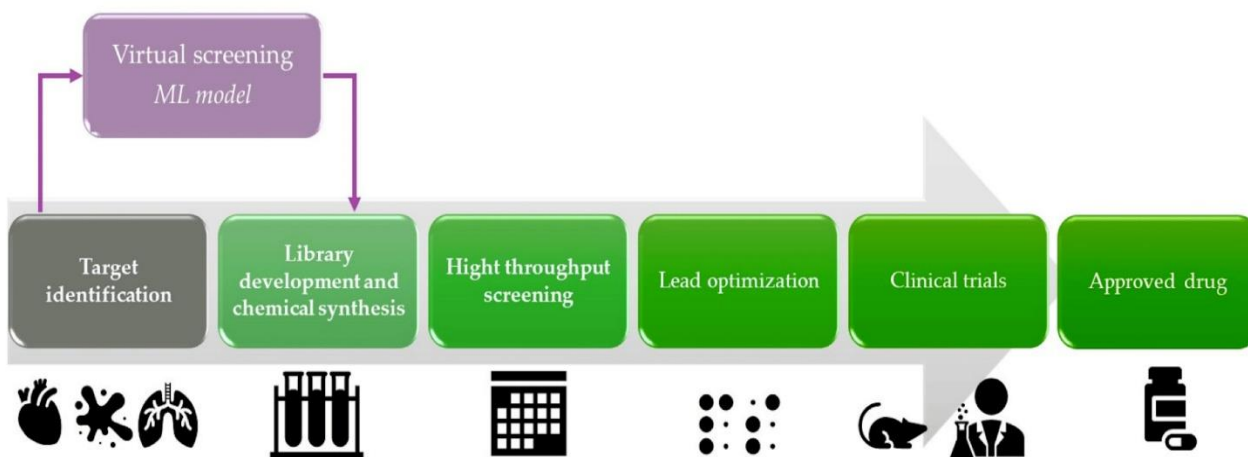


Figure 3. A machine learning model is used in the drug development process to do virtual screening.

Artificial intelligence approaches, such as deep learning and quantum chemistry, can be employed to do predictive modeling and optimization in the field of drug creation. Deep learning models have the ability to examine large quantities of data, such as chemical structures, biological activities, and pharmacokinetic parameters, in order to make predictions about the effectiveness and safety of proposed medications [26]. The integration of quantum chemistry with artificial intelligence (AI) can expedite the investigation of chemical space and provide guidance for the development of novel therapeutic candidates [25]. It is crucial to acknowledge that although AI has demonstrated encouraging outcomes in virtual screening and drug creation, experimental validation remains imperative for the identified therapeutic candidates. Artificial intelligence (AI) is a potent technology that can aid researchers in the initial phases of drug development. However, it is crucial

to supplement AI with conventional experimental techniques and thorough testing [23]. In general, artificial intelligence (AI) has the capacity to greatly speed up the drug discovery process, enhance the effectiveness of virtual screening, and facilitate the creation of new drug candidates with improved attributes. With the ongoing progress in AI and ML research, we can anticipate future enhancements in virtual screening and drug creation approaches.

(iii) ***Drug Property Prediction***

Artificial intelligence systems have the capability to forecast the physicochemical characteristics of potential drugs, including their solubility, bioavailability, and toxicity [16]. This approach aids in the optimization of medication development by prioritizing compounds with a greater probability of success, hence diminishing expenses and time requirements. Machine learning algorithms can be trained to forecast pharmacological characteristics by utilizing a database of established compounds. These algorithms acquire the ability to identify patterns and connections between the chemical composition of a substance and its physical and chemical characteristics [20]. After undergoing training, these models have the ability to forecast the physicochemical characteristics of untested substances, which can be helpful in the first phases of drug development [20]. ML models have been employed to forecast solubility, a vital characteristic for potential therapeutic candidates [27]. Furthermore, machine learning algorithms have been employed to forecast the bioavailability of medication candidates, which is also a crucial characteristic [28]. AI has demonstrated potential in the field of toxicity prediction. Machine learning methods have been employed to forecast the toxicity of substances by analyzing their chemical structures [29]. These models aid in the initial stages of drug discovery by finding molecules with a lower likelihood of toxicity, thereby minimizing the potential harm to humans during the testing phase [14].

(iv) ***Drug repurposing***

Artificial intelligence allows researchers to discover novel therapeutic uses for current medications through the analysis of extensive databases containing information on drug-target interactions and illness pathways. Utilizing this strategy of repurposing drugs might greatly minimize the duration and expenses involved in bringing a medicine to the market (Figure 4) [30]. The application of AI in medication combination identification involves improved detection of synergy, optimization of combinations, tailored combination therapy, and prediction of adverse drug interactions.

Artificial intelligence algorithms examine complex biological data to detect possible synergistic combinations of drugs. Through the examination of interactions between medications and their targets, artificial intelligence (AI) has the ability to forecast combinations that demonstrate improved therapeutic outcomes while reducing side reactions [31]. Artificial intelligence has the capability to enhance the effectiveness of drugs and minimize their negative effects by optimizing the dosage and timing of administration in combination therapies [32]. By fine-tuning, the complete therapeutic effects of the combination are ensured, hence increasing its viability as a therapy choice. AI-powered precision medicine enables the identification of personalized drug combinations for patients, using their unique molecular profiles and illness characteristics [33].

The objective of this individualized strategy is to attain superior therapeutic results and reduce the likelihood of developing drug resistance. AI models have the ability to forecast any negative interactions between pharmaceuticals when taken together, therefore protecting the safety of patients and preventing potential harm [34].

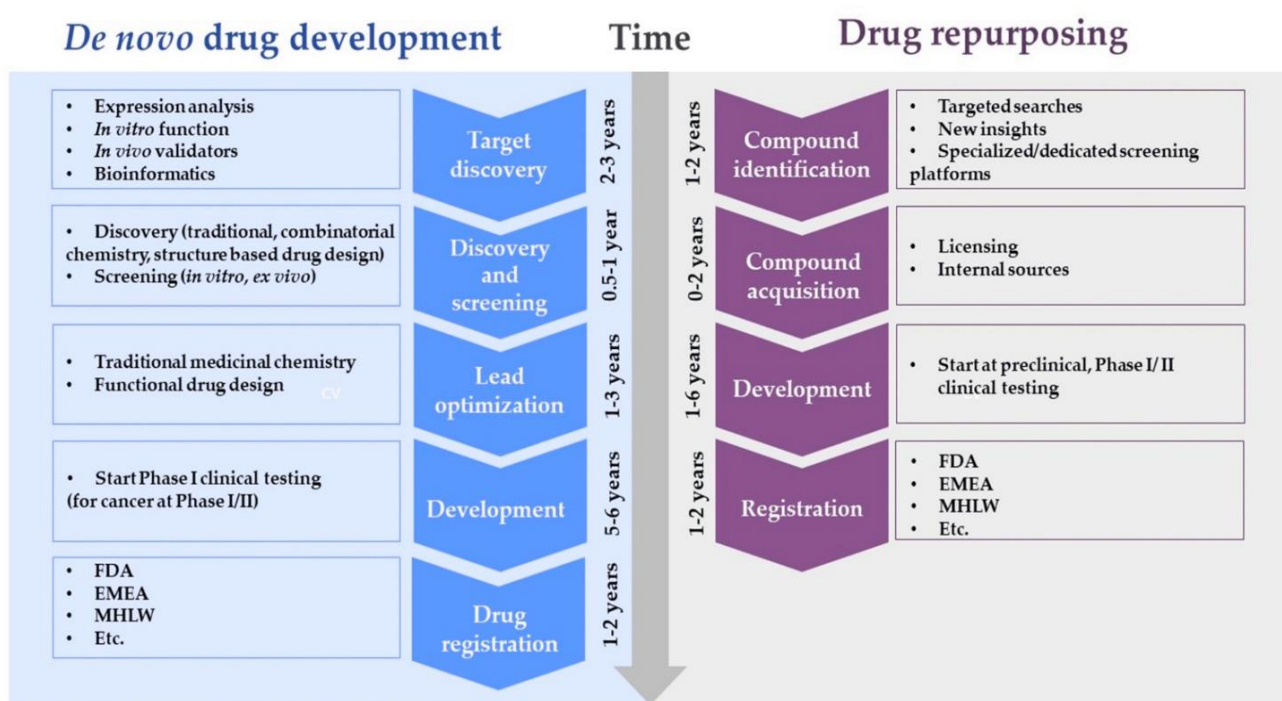


Figure 4. Utilizing a drug repurposing strategy.

2.1. Artificial intelligence for material discovery:

The primary objective of AI approaches, specifically in the field of materials science, is to uncover and create novel materials that possess highly desirable characteristics. This is highly relevant to drug discovery, as the procedure frequently necessitates the identification of new molecules with precise biological activity. Artificial intelligence algorithms that demonstrate exceptional performance in forecasting the characteristics of novel materials can also accurately forecast the pharmacological profiles of potential drug candidates. This integration enables the more effective evaluation and enhancement of new medications, utilizing the predictive capacities of artificial intelligence to simplify both the material and drug discovery procedures.

Artificial intelligence methods used in materials discovery encompass both supervised and unsupervised learning approaches. Supervised learning is a machine learning approach that utilizes data with known labels to train models capable of classifying or predicting outcomes for fresh data. Unsupervised learning involves working with data that does not have labels and seeks to create models that can recognize repeated patterns and groups within the data without any prior information. On the other hand, in the field of drug discovery, artificial intelligence (AI) approaches are employed at several points, such as gathering and organizing data, representing compounds, and applying AI methodologies. Data resources, data representation schemes, and AI methodologies are the three fundamental elements required for the implementation of AI in the fields of drug development and evaluation. AI techniques employed in drug discovery encompass regression analysis, decision trees, logistic regression, support vector machines, convolutional neural networks, recurrent neural networks, generative adversarial networks, k-means clustering, hierarchical clustering, principal component analysis, and t-distributed stochastic neighbor embedding.

When it comes to drug development, AI techniques are carefully chosen and applied in a way that is tailored to the specific problems at hand, in order to assure optimal results. Supervised and unsupervised learning are two often employed AI strategies in the field of drug development [35].

Table 1. Commonly employed AI techniques in the field of pharmaceutical research.

Class	Algorithms	Reference
<u>Supervised learning</u>	Regression analysis	<u>MLR</u> [35]
		<u>DT</u> [36]
		<u>LR</u> [37]
	Classification	<u>SVM</u> [38]
		<u>CNN</u> [39]
		<u>RNN</u> [40]
		<u>GAN</u> [41]
<u>Unsupervised learning</u>	Clustering	<u>k-means</u> [42]
		Hierarchical [43]
	Dimensionality reduction	PCA [44]
		t-SNE [45]

2.1.1. Methods for supervised learning

Supervised learning is a process where models are trained using data that has been labeled with inputs. These models are then able to categorize or predict outcomes for fresh data. For example, the output could be a property such as band gap or thermal conductivity. [46] Supervised learning models have the ability to make predictions about the characteristics of new materials by utilizing the patterns they have learnt during the training process [47]. Unsupervised learning, in contrast, focuses on analyzing unlabeled data to develop models that can detect repeating patterns and clusters in the input data without any prior knowledge [48]. They can assist in identifying new material categories that may not be evident through manual analysis [49]. Supervised learning techniques can be divided into two categories: classification algorithms and regression algorithms. On the other hand, unsupervised learning techniques encompass clustering algorithms and dimensionality reduction algorithms.

Linear regression is a statistical technique that models the connection between a dependent variable and one or more independent variables by using a linear equation to analyze the available data. Linear regression is a useful tool in drug development for predicting a compound's physicochemical quality by analyzing its molecular descriptors. Ridge regression and lasso regression are variants of linear regression that include shrinkage, a method for lowering model complexity. Support Vector Regression (SVR) is a regression analysis technique that utilizes the concept of support vectors. The approach is versatile for regression and can handle both linear and non-linear situations. Additionally, it has the capability to process data with a large number of dimensions and is highly efficient in scenarios with significant levels of noise. Reinforcement learning (RL) can be used to optimize the synthesis process in material discovery [50]. Reinforcement Learning (RL) is a subfield of Machine Learning (ML) in which an autonomous agent acquires the ability to make optimal judgments by taking actions within an environment in order to attain specific objectives. The agent is provided with feedback in the form of rewards or penalties, which helps it to acquire

the optimal strategy, referred to as a policy. Reinforcement learning is a process that incorporates experimentation, where the agent tries different behaviors and learns from the results in order to improve its behavior. This approach is especially valuable in situations where there are no clear instructions on how to attain a goal. It enables the agent to independently discover appropriate solutions through a process of iterative learning and adaptation. This method is becoming increasingly popular in many industries, including as robotics, gaming, and autonomous cars, because of its capacity to adjust and enhance based on experience [50]. Q-learning and deep Q-Networks (DQNs) are two frequently used reinforcement learning (RL) methods. Q-learning is a model-free algorithm used in reinforcement learning. It is capable of determining the value of an action in a specific state without the need for an environment model. A Deep Q-Network (DQN) is a hybrid model that combines deep learning and Q-learning, enabling the agent to learn directly from unprocessed visual data or other unprocessed input. One of the primary benefits of RL is its holistic approach, addressing the problem as a unified entity rather than breaking it down into smaller components. The system has the ability to prioritize long-term advantages above immediate gains, and it does not need to gather data separately because it acquires training data through direct interaction between the agent and the environment. In order to simplify the implementation of these artificial intelligence techniques, several open-source packages and frameworks have been created, including Scikit-learn [51], PyTorch [52], and Keras [53]. These packages offer consumers the opportunity to utilize a variety of algorithms in the field of drug discovery. Regression analysis techniques are essential in drug development, specifically for predicting the physicochemical features of substances. These algorithms can enhance the drug development process by discovering compounds with favorable features and minimizing the time and cost of testing.

Regression analysis is a statistical technique employed in drug development to forecast the result of an experiment by considering the values of predictor variables [54]. Regarding drug discovery, the result may refer to a physicochemical characteristic of a substance, such as its solubility, bioavailability, or toxicity [54]. Linear regression is a fundamental and widely employed approach for doing regression analysis. The process involves fitting a linear equation to the observed data in order to represent the connection between a dependent variable and one or more independent variables [55]. Linear regression is a useful tool in drug development for predicting the physicochemical quality of a substance using its molecular descriptors [56]. Ridge regression is a form of linear regression that employs shrinkage, a method for decreasing the complexity of a model [57]. The shrinkage parameter, also known as the tuning constant, controls the extent of shrinkage. A higher value of the shrinkage parameter corresponds to a bigger degree of shrinkage [58]. Lasso regression is a variant of linear regression that incorporates shrinkage and is capable of handling high-dimensional data [59]. The primary distinction between ridge and lasso regression lies in the fact that lasso regression can yield sparse solutions, wherein certain feature coefficients are precisely zero, thus removing those features from the model [60]. Support Vector Regression (SVR) is a regression analysis technique that incorporates the concept of support vectors [38]. The approach is versatile for regression and can effectively handle both linear and non-linear situations. Additionally, it has the capability to process data with a large number of dimensions and is highly efficient in scenarios with significant levels of noise [38].

Classification is the process of categorizing data elements into several predefined groups. Classification algorithms are utilized in drug discovery to forecast the efficacy of a molecule.

A support vector machine (SVM) is a supervised learning method utilized for classification and regression analysis. The algorithm operates by identifying a hyperplane within a space with several dimensions that effectively separates the data points into discrete classes. SVM, or Support Vector

Machines, can be employed in the field of drug development to categorize compounds according to their attributes and forecast their efficacy.

A convolutional neural network (CNN) is a deep learning technique specifically designed for image processing. Additionally, it can be employed in the field of drug development to perform tasks such as forecasting the characteristics of novel compounds. A Convolutional Neural Network (CNN) possesses the ability to autonomously acquire and isolate characteristics from unprocessed data, rendering it well-suited for managing intricate and multi-dimensional data.

A recurrent neural network (RNN) is a deep learning technique specifically designed for handling sequential input. For drug development purposes, an RNN can be employed to forecast the characteristics of novel compounds by analyzing their sequence data.

A generative adversarial network (GAN) is a deep learning system including two neural networks, namely the generator and the discriminator. These networks are trained in an adversarial way to produce authentic data samples. GANs are utilized in the field of drug development to generate novel molecular structures that possess specific features.

2.1.2. Methods for unsupervised learning

Unsupervised learning entails utilizing data that lacks labels to discern patterns and correlations inherent in the data. Unsupervised learning approaches in drug discovery can be employed to discover new material classes or to investigate the chemical space of possible therapeutic candidates. Clustering, specifically k-means, is an unsupervised learning method that categorizes data points by their likeness. Clustering is a useful technique in drug development for identifying sets of molecules that share similar characteristics. This can aid in the identification of new and promising therapeutic candidates. Hierarchical clustering is a clustering technique that constructs a hierarchy of clusters using either a bottom-up or top-down approach. Hierarchical clustering is a useful method in drug discovery for identifying groupings of molecules that share similar features at various degrees of detail. Dimensionality reduction techniques such as PCA and t-SNE are employed to decrease the number of variables in a dataset while preserving its fundamental properties. Principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE) are widely used methods for reducing the dimensionality of data. Dimensionality reduction is a technique employed in drug development to visually represent complex data sets with many variables, enabling the identification of hidden patterns or trends that may not be readily observable in the original data.

2.2. Artificial intelligence algorithms for drug discovery:

Traditional methods in drug discovery mainly depend on manual investigation, experimentation, and testing. In contrast, AI-driven methodologies utilize data-driven analysis, machine learning, and rapid simulation.

Machine learning and deep learning

Machine Learning (ML) and Deep Learning (DL) are specific branches of Artificial Intelligence (AI) that have been successfully utilized in the field of drug discovery. Although there are certain commonalities between them, they exhibit clear disparities in terms of their strategy, architecture, and capabilities [61].

(i) ***Artificial Intelligence***

Machine learning is a vast domain that comprises a range of algorithms capable of identifying patterns and making predictions from data. These algorithms usually operate on organized data and necessitate feature engineering, which involves selecting or creating pertinent characteristics prior to inputting the data into the model [61,62,63]. Feature engineering is an essential and crucial process in preparing the data for machine learning models [63]. Conventional machine learning techniques necessitate structured and labeled data for the purpose of training. Machine learning techniques are typically less complex and demand lower computational resources in comparison to deep learning models [61,62]. While they may excel in specific tasks with a restricted amount of data, they may encounter difficulties when faced with intricate and non-linear problems. Conventional machine learning models sometimes necessitate a significant quantity of labeled data in order to attain satisfactory performance, particularly in intricate jobs. Conventional machine learning models are typically more comprehensible, indicating that it is simpler to comprehend the reasoning behind the model's predictions by considering the chosen features and parameters [61,62].

(ii) ***Deep Learning***

Artificial neural networks that are capable of learning and making complex decisions based on large amounts of data. However, DL is a specific type of ML that use artificial neural networks to acquire knowledge about data representations [64]. Neural networks possess numerous layers, enabling them to acquire hierarchical properties from unprocessed data. Unlike conventional machine learning (ML) methods, deep learning (DL) models have the ability to autonomously acquire features from data, hence removing the necessity for extensive feature engineering [65]. Deep learning models have the ability to process unstructured input, such as photos, phrases, and sequences, without requiring complex feature engineering. Neural networks acquire hierarchies of representations directly from unprocessed data, which makes them more equipped to handle intricate and multi-dimensional data [65]. Deep learning models are characterized by their intricate nature and demand substantial computer resources, particularly when handling extensive datasets. Artificial intelligence models are highly proficient in managing intricate patterns and non-linear connections in data, which makes them especially well-suited for tasks such as image and language processing [65]. Deep learning models frequently exhibit superior performance with a reduced amount of labeled data, owing to their capacity to autonomously acquire features and representations from the data. Deep learning models are commonly seen as less interpretable because of their intricate architecture with several layers and sophisticated learnt representations. Comprehending the decision-making process of deep learning models can be arduous. Gong D. In their study, et al. showcased the application of machine learning technologies in silico to evaluate polymers for gene delivery [66]. This study utilized advanced machine learning techniques to analyze a dataset of synthetic biodegradable polymers, specifically PBAEs. These polymers have demonstrated potential for delivering therapeutic genes both in laboratory settings (in vitro) and in living organisms (in vivo) [66]. The dataset comprised polymer characteristics as inputs, together with polymeric nanoparticle transfection performance and nanoparticle toxicity in various cells as outputs. The data were utilized to train and assess multiple cutting-edge machine learning algorithms in order to forecast transfection and comprehend structure-function correlations [66]. The authors successfully demonstrated that by creating an encoding scheme to convert the structure of a PBAE polymer into a machine-readable format, a random forest model can accurately predict DNA transfection in vitro. This prediction is based on the chemical structure of the PBAE polymer and is dependent on the specific cell line used [66]. Therefore, a computational method that

represented the chemical characteristics of polymers showed that using computer simulations to screen polymeric nanomedicine compositions was useful in predicting new biological trials.

Deep Learning (DL) is a subset of Artificial Intelligence (AI) that utilizes both supervised and unsupervised learning methods, depending on the specific problem and data being examined [67].

(iii) ***High-throughput Density Functional Theory (DFT)***

High-throughput Density Functional Theory (DFT) is a computational method that allows for the efficient and rapid calculation of electronic properties of materials.

High-throughput density functional theory (DFT) simulations are computationally demanding yet essential for comprehending material characteristics at the atomic scale [68]. High-throughput Density Functional Theory (DFT) utilizes optimized algorithms and parallel processing to quickly forecast the characteristics of many materials [68]. The application of DFT in drug development has gained significance due to its pivotal role in comprehending and forecasting the interactions between therapeutic compounds and biological targets.

High-throughput Density Functional Theory (DFT) can be employed to compute the binding energies and binding affinities between drug compounds and their target proteins [69]. These calculations aid in the identification of possible therapeutic candidates that exhibit robust binding interactions, hence facilitating a more streamlined process of screening and selecting promising molecules. High-throughput Density Functional Theory (DFT) can be employed in virtual screening to forecast the binding affinities of an extensive collection of drug-like molecules towards a particular target [70]. Through the process of virtual screening, researchers are able to efficiently evaluate a vast number of substances and identify the most favorable candidates for subsequent experimental confirmation. Researchers can utilize high-throughput DFT to examine the molecular-level characteristics of drug-target interactions [71]. The process can determine the specific amino acid residues that are crucial for binding and assess how alterations to the ligand affect the strength of the binding. This information is valuable for the strategic design of potent and specific medications [72]. The high-throughput DFT algorithm can accurately forecast physicochemical parameters, including solubility, lipophilicity, and metabolic stability. These features are essential for evaluating the pharmacokinetics and toxicity of a potential drug candidate [73]. These forecasts assist in choosing compounds with advantageous ADMET profiles and decrease the probability of failure in subsequent phases of drug development. Additionally, high-throughput Density Functional Theory (DFT) can be employed to investigate drug metabolisms by forecasting the energy levels of different metabolic events, such as hydroxylation or oxidation [74,75]. These predictions aid in the identification of prospective metabolic sites and enhance the comprehension of the metabolic pathways of medication candidates. Alternatively, high-throughput Density Functional Theory (DFT) can analyze the flexibility of ligands and determine the most stable conformers [76]. Accurate representation of ligand flexibility is crucial in molecular docking and molecular dynamics simulations, since it enhances the reliability of binding predictions. Fragment-based drug design can utilize high-throughput density functional theory (DFT) to examine the interaction between tiny fragments and a specific protein target. These calculations aid in the identification of fragments that can be combined to form larger, more potent molecules with drug-like properties [77].

(iv) ***Natural Language Processing (NLP)***

Natural Language Processing (NLP) is the field of study that focuses on the interaction between computers and human language. It involves developing algorithms and models to enable

computers to understand, interpret, and generate human language in a way that is both accurate and efficient.

Natural Language Processing (NLP) is a crucial artificial intelligence approach employed in the field of text mining for the purpose of drug discovery. Natural Language Processing (NLP) algorithms have the ability to analyze and comprehend human language in order to extract pertinent information from various textual sources such as scientific publications, patents, clinical trial data, and more [78]. Natural Language Processing (NLP) enabled artificial intelligence (AI) models have the capability to recognize drug names, target proteins, chemical substances, and disease-related information. This ability assists researchers in collecting vital data for the purpose of drug discovery [78].

(v) ***Text mining***

Text mining refers to the process of extracting valuable information or knowledge from large amounts of text data. Text mining allows for the discovery of already existing medications that could be potentially used in various therapeutic fields [79]. Through literature analysis, AI models can propose potential drugs that have proven effectiveness against specific diseases or targets. This opens up possibilities for repurposing drugs and investigating the concept of polypharmacology [79]. Artificial intelligence models have the capability to combine many sources of data, including clinical trial outcomes, genomic information, and chemical databases, in order to construct knowledge graphs [80]. Knowledge graphs are graphical representations that capture intricate connections between medications, targets, diseases, and biological pathways. They enable thorough research and the formulation of hypotheses [80]. Additionally, AI models have the capability to extract information from scientific literature in order to identify adverse medication reactions that have been documented in both clinical studies and post-marketing surveillance. An important aspect of enhancing drug safety evaluations and informing decision-making in clinical trials is the extraction of adverse drug reaction information from scientific literature [81]. AI algorithms can utilize text mining to discover potential biomarkers linked to specific diseases or pharmacological reactions [82]. Biomarkers are essential in customized medicine as they help categorize patients and facilitate the creation of specific treatments.

AI-driven strategies have been used to harness various text mining-based tools and maximize their capabilities. Jang et al. developed PISTON, a tool that utilizes natural language processing (NLP) and topic modeling to forecast pharmacological side effects and indications [83]. The tool can be accessible at <http://databio.gachon.ac.kr/tools/PISTON/> (accessed on 1 February 2024). DisGeNET is a database that uses text mining to provide extensive information on links between genes and diseases, as well as variants and diseases. It may be visited at <https://www.disgenet.org/> (accessed on 1 February 2024) [84]. DisGeNET's data analysis covers a wide range of biological processes, such as adverse drug reactions, molecular pathways related to diseases, and the effects of drugs on targets. Another resource available is STRING, a text mining-driven database accessible at <https://string-db.org/> (accessed on 1 February 2024). This resource provides a wealth of data on protein-protein interactions in different animals [85]. Furthermore, the website STITCH (<http://stitch.embl.de/> (accessed on 1 February 2024)) offers significant insights into the interactions between proteins and tiny molecules, such as chemicals [86]. The information from STITCH is also employed to ascertain the affinities of drug binding and the relationships between drugs and targets.

(vi) ***Generative Adversarial Networks (GANs)***

GANs, short for *Generative Adversarial Networks*, are a type of machine learning techniques that involve two neural networks: the generator and the discriminator. These networks

are trained in an adversarial fashion, meaning they compete against each other, to produce realistic data samples. [87] GANs are utilized in drug discovery for the purpose of generating novel chemical structures [88] and enhancing lead compounds [89], among other uses.

For instance, Generative Adversarial Networks (GANs) have the capability to produce unique chemical structures that possess specific desirable characteristics. The generator network acquires the ability to generate chemically accurate structures, while the discriminator network assesses their credibility by comparing them to a dataset used for training. GANs are valuable in de novo drug discovery because they stimulate the generator to create novel compounds that possess similar characteristics to existing medications [90].

Furthermore, it has been discovered that Generative Adversarial Networks (GANs) can be utilized to enhance the efficiency of lead compounds by producing alterations or analogues of current medication candidates. GANs utilize the discriminator's feedback to direct the synthesis of novel chemical structures, resulting in proposed alterations that are more probable to exhibit biological activity and possess enhanced pharmacological qualities [91].

Several research have demonstrated that GANs can be utilized for multi-objective drug design, aiming to optimize numerous pharmacological attributes concurrently [41]. Researchers may manipulate the loss function of the GAN to achieve a balance between several parameters, such as binding affinity, solubility, and selectivity. This allows them to create molecules with desired multi-objective profiles [41,89]. Prior research [92] has demonstrated that GANs have the capability to optimize distinct molecular characteristics, such as lipophilicity, molecular weight, or solubility. Researchers can use this skill to precisely adjust lead compounds or create molecules with certain physicochemical features [93].

Based on the current research, it has been established that GANs have the ability to investigate uncommon chemical compositions and detect chemicals that are not often present in conventional drug databases. This capability is highly advantageous for identifying new chemical compounds with distinct characteristics, which could ultimately result in the creation of groundbreaking pharmaceuticals [94].

Research indicates that Generative Adversarial Networks (GANs) can be incorporated into virtual screening pipelines to provide innovative medication ideas. GANs can enhance the chemical diversity of virtual screening libraries and improve the likelihood of finding hits against specific targets by producing a wide range of compounds [95].

GANs have the ability to enhance small datasets of confirmed active substances, which leads to improved generalization and enhanced performance of prediction models by generating more data samples [88]. Furthermore, GANs improve the effectiveness and precision of the machine learning models employed in the process of drug discovery [96].

Contemporary scientific research suggests that Generative Adversarial Networks (GANs) have the potential to forecast drug-target interactions by producing chemical structures that are probable to interact with particular protein targets. Subsequently, the synthesized compounds might be empirically confirmed to ascertain potential new interactions between drugs and targets [94,97].

(vii) ***Transfer learning***

Transfer Learning enables the adjustment of pre-trained models, which were initially trained on a larger dataset, to specialize on a certain material class or property using a smaller dataset. This is particularly advantageous in situations where there is a limited supply of annotated

data. Transfer learning in drug development is a machine learning technique that utilizes knowledge acquired from one activity and applies it to another similar task with a small amount of data. This method has demonstrated significant potential in tackling the issue of sparsely labeled data in *in silico* drug discovery endeavors [98]. Transfer learning involves utilizing a pre-trained model that was first constructed for a specific task, such as image recognition. This pre-trained model serves as the foundation for a new model that is designed to address a different goal, such as predicting drug-target interactions or discovering pharmacological compounds with high efficacy [99]. By leveraging the acquired representations from the pre-trained model, the new model can exploit the knowledge and generalization skills of the pre-trained model, even when provided with a limited quantity of labeled data. Previously, quantitative structure–activity relationship (QSAR) models employed regression models to establish connections between molecular descriptors and biological features [100].

In drug development, machine learning approaches like support vector machine algorithms and decision trees have been used to classify drugs and predict their absorption, distribution, metabolism, excretion, and toxicity (ADME/T) features [101,102,103]. AI technology has achieved significant advancements in several domains, such as predicting molecular characteristics and activities [104,105,106], virtual screening [107,108], retrosynthetic analysis [109,110], and producing novel pharmaceuticals [111,112]. Deep transfer learning, utilizing deep neural networks, is the predominant kind of transfer learning employed in drug discovery [113]. Convolutional neural networks (CNN) and recurrent neural networks (RNN) are advanced machine learning models that have been used in many drug discovery activities, such as predicting compound activity, doing virtual screening, and creating new drugs [113]. These models possess the ability to acquire intricate representations of chemical structures or biological facts and apply that knowledge to similar drug discovery activities. Deep learning employs deep neural networks with several hidden layers to comprehend and interpret intricate information, as opposed to conventional "shallow" machine learning methods. Big data is a term used to describe datasets that are large in size, come from different sources, are updated quickly, include all the features of a system, are valuable in terms of insights and potential for reuse, and have other specific characteristics [114]. Transfer learning provides a notable benefit in the field of drug development by addressing the obstacle of having limited and heterogeneous datasets. Researchers can utilize pre-trained models to extract valuable information from extensive and diverse datasets in relevant domains [98]. For instance, a pre-trained Convolutional Neural Network (CNN) model, which has undergone training on a vast collection of images, can be adjusted and utilized to examine chemical structures or molecular fingerprints. This allows for the prediction of compound activity or the identification of probable drug-target interactions [98]. Transfer learning can expedite the discovery of new medications and improve the accuracy of finding powerful medicinal molecules. By leveraging the information held in pre-existing models, researchers can reduce the time and resources required to train models from scratch [98]. Utilizing transfer learning in drug discovery expedites testing and experimentation, resulting in rapid identification of interesting drug candidates. However, it is crucial to remember that transfer learning in the field of drug discovery remains a subject of continuous research, with many obstacles and constraints that need to be taken into account. One of the difficulties is in the process of choosing the appropriate pre-trained models and subsequently refining them accurately. When selecting a pre-trained model, it is essential to take into account its pertinence to the desired job and the accessibility of suitable datasets [35]. It should be emphasized that transfer learning does not consistently result in improved performance. This is especially accurate when the source and target tasks exhibit significant dissimilarities. It is essential to thoroughly evaluate the degree to which knowledge may be applied to different tasks in order to

ensure that the transferred information is pertinent and valuable for the intended target activity [98]. In summary, transfer learning is a promising approach in the field of drug discovery that addresses the issue of limited labeled data and accelerates the identification of potent therapeutic molecules. The technique entails utilizing pre-trained models and leveraging expertise from analogous jobs to augment the drug discovery process. This approach can assist researchers in acquiring valuable representations and enhancing the efficiency of the procedure. Nevertheless, further study is required to enhance the process of selecting and refining pre-trained models, as well as to investigate the wider potential of transfer learning in the field of drug development.

(viii) ***Active learning***

Active learning can effectively reduce the number of experiments needed by actively selecting useful samples to test or simulate. The application of artificial intelligence (AI) methods in the field of material discovery has resulted in remarkable progress in the field of drug discovery. Researchers can optimize the design and discovery of materials for many purposes by utilizing computational power and data-driven insights. This consequently speeds up the process of creating new and improved ideas and technologies in various sectors [98]. Active learning is a machine learning technique that is employed in the field of drug development. The objective is to enhance the process of choosing compounds for experimental testing by repeatedly selecting the most informative samples to label. This method minimizes the requirement for extensive screening of extensive compound libraries and instead concentrates on specific regions of chemical space that have the greatest likelihood of success, taking into account structural originality as well [115]. Active learning is a process where a small set of compounds with known labels is first selected, and further compounds are chosen for labeling depending on the predictions made by the model. After the compounds are chosen, they undergo testing. The results obtained from the testing are then utilized to enhance the model and refine the selection process for subsequent cycles [115]. Active learning algorithms may effectively navigate and utilize the chemical space by deliberately selecting which compounds to label. This leads to more efficient and cost-effective drug discovery methods. These algorithms aim to achieve higher predicted accuracy while employing a smaller number of labeled examples compared to standard passive learning approaches [115]. Active learning strategies can be advantageous for numerous drug discovery activities. These activities encompass chemical activity prediction, lead optimization, and virtual screening. Active learning is applicable to multiple machine learning models, including Support Vector Machines (SVM), Random Forest (RF), and neural networks [115]. Active learning in drug discovery provides numerous advantages, such as cost and resource savings. This is accomplished by choosing a specific group of chemicals to examine in experiments, which reduces the necessity for thorough screening and saves time and money [115]. Active learning algorithms can effectively navigate chemical space by selecting substances that are highly likely to provide valuable information. This methodology results in a more focused and efficient process for discovering drugs [115]. Regularly updating the model with experimental data can improve its predicted accuracy, which is known as active learning. Over time, this method results in more accurate forecasts of chemical activity [115]. When using active learning in drug discovery, several obstacles and factors must be taken into account. One of the most significant challenges is selecting the appropriate machine learning model and fine-tuning its parameters, as this is essential for achieving success. It aligned with the active learning process [115]. It is essential to carefully choose the models, data selection methodologies, and workflow integration in order to effectively include active learning into drug discovery pipelines. we have compiled a comprehensive list in Table 2 of collaborations between pharmaceutical corporations and AI providers.

Table 2. Partnerships between pharmaceutical businesses and AI suppliers

Pharmaceutical Company	AI Provider	Project Details
AstraZeneca	BenevolentAI	Identified the first AI-generated drug target for chronic kidney disease (CKD).
Pfizer	AI Technology	Utilized AI for COVID-19 vaccine trials and distribution. Additionally, applied AI and predictive analytics to modernize and streamline drug development.
Pfizer	XtalPi	Developed a hybrid platform combining physics and AI for precise molecular modeling of drug-like small molecules.
Pfizer	Insilico Medicine	Leveraged data mining for drug target identification.
Pfizer	ConcertAI	Enhanced study design and diversified clinical trials through expanded AI partnership.
Pfizer	Janssen Research and Development (Johnson & Johnson)	Applied AI for identifying and selecting new targets and disease subsets to support therapeutic programs.
Eli Lilly, Bayer, Bridge Biotherapeutics	Atomwise	Assisted in structure-based small-molecule drug discovery.
Boehringer Ingelheim	Google Quantum AI	Collaborated to use quantum computing for accelerating and optimizing new medicine discovery.
Boehringer Ingelheim	Insilico Medicine	Partnered to use AI for identifying potential therapeutic targets.
BMS	Exscientia	Selected an AI-designed immune-modulating drug candidate through a contractual agreement.
GSK	Exscientia	Developed the first AI-powered treatment for COPD.
Roche, Sanofi, Bayer	Exscientia	Collaborated with these pharmaceutical companies.
AstraZeneca	Eko, BERG, Renalytix AI, Mila-Quebec AI Institute	Employed AI algorithms and supercomputers for drug discovery.
Optellum	J&J Lung Cancer Initiative	Used an AI-powered clinical decision support platform to improve early lung cancer treatment.

3. Research on the application of AI in drug discovery.

Artificial Intelligence (AI) provides a potent tool for drug discovery, encompassing various applications such as forecasting drug characteristics, creating novel chemical structures, and enhancing drug properties (Figure 5). ML algorithms are frequently used in drug discovery as a prominent application of AI. These algorithms have the capability to acquire knowledge from extensive collections of chemical structures and biological data in order to forecast the characteristics of novel compounds, such as their capacity to attach to a certain protein (affinity prediction) or their potential for causing harm (toxicity prediction). This can aid in the identification of prospective medication candidates that can undergo laboratory testing. AI can also be applied in drug discovery by utilizing Generative Adversarial Networks (GANs) to produce novel chemical structures that have a high probability of binding to a specific protein. This is achieved by analyzing the chemical structures of existing active and inactive molecules. AI can be utilized in the field of drug development to do multi-objective optimization. This involves optimizing a medicine to achieve many desirable features, such as high efficacy, low toxicity, and good solubility.

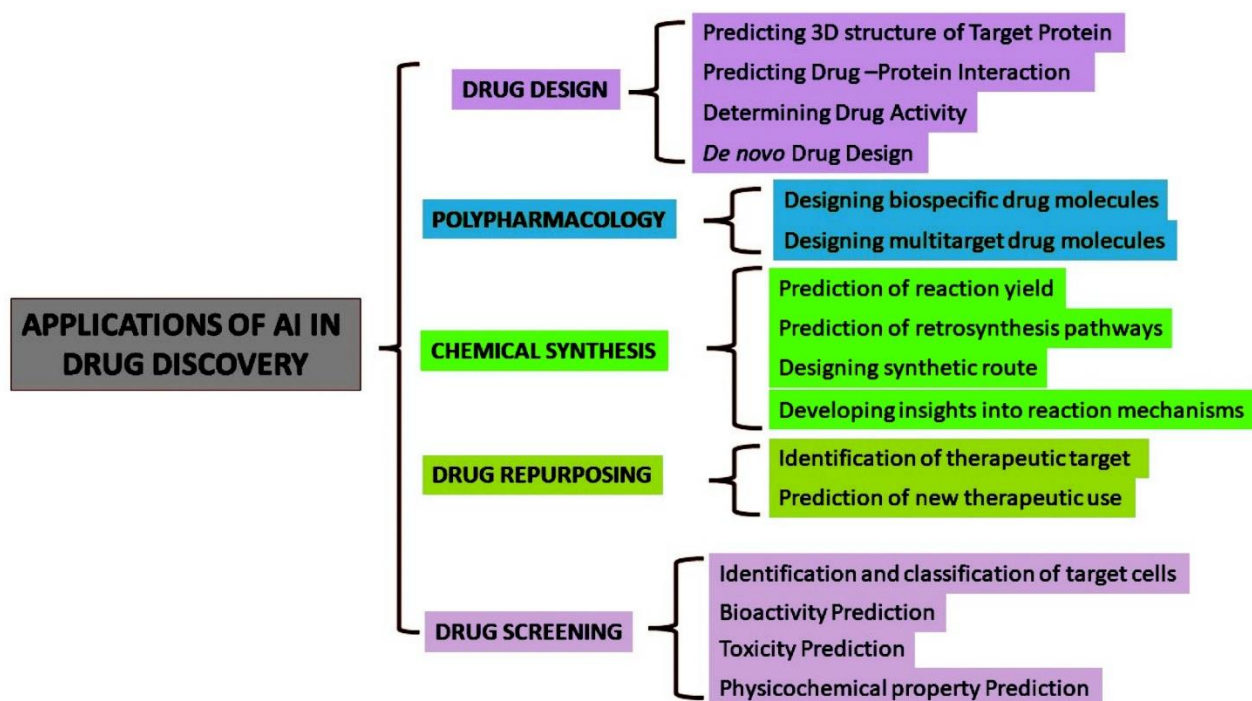


Figure 5. Utilizations of artificial intelligence in the field of drug research.

3.1. Artificial intelligence (AI) programs and platforms are utilized in the field of drug discovery.

Various artificial intelligence techniques and platforms are currently employed in the field of drug discovery, as indicated in Table 3.

Table 3. Artificial intelligence techniques and platforms are currently being utilized in the field of drug research.

Program	Description	Primary Use	Accession
DeepChem	Python-based AI system using MLP model	Candidate selection in drug discovery	https://github.com/deepchem/deepchem (accessed on 20 June 2024)
DeepNeuralNetQSAR	Python-based AI system	Can aid the detection of the molecular activity of compounds	https://github.com/Merck/DeepNeuralNet-QSAR (accessed on 20 June 2024)
Chemputer	Combination of Monte Carlo tree search and symbolic AI, including DNNs	Synthesize organic molecules	https://zenodo.org/record/1481731 (accessed on 20 June 2024)
DeepTox	AI system using DL	Chemical toxicity prediction	http://www.bioinf.jku.at/research/DeepTox (accessed on 20 June 2024)
AlphaFold	AI system using DL	Predicts the 3D structures of proteins	https://alphafold.ebi.ac.uk/ (accessed on 20 June 2024)
ORGANIC	Generative ML approaches and DNNs	Novel molecular materials	https://github.com/aspuru-guzik-group/ORGANIC (accessed on 20 June 2024)
PotentialNet	Neural networks, deep attention mechanisms and descriptor embeddings	The binding affinity of ligands in protein–ligand complexes	https://www.genesistherapeutics.ai/platform.html (accessed on 20 June 2024)
Hit Dexter	ML technique, CNNs and ANNs	For predicting molecules that might respond to biochemical assays	http://hitdexter2.zbh.uni-hamburg.de (accessed on 20 June 2024)
DeltaVina	ML algorithms, including XGBoost and random forest	Scoring protein–ligand binding affinity	https://github.com/chengwang88/deltavina (accessed on 20 June 2024)
Neural graph fingerprint	CNNs	Predict properties of novel molecules	https://github.com/HIPS/neural-fingerprint (accessed on 20 June 2024)
GastroPlus	AI and predictive modeling	For pharmaceutical products (dosage form) in many animal models	https://www.simulations-plus.com/software/gastroplus/# (accessed on 20 June 2024)

As an illustration, an MLP model utilizes a Python-based AI system called **DeepChem** to identify an appropriate candidate in the field of drug development. This model may be viewed at [116].

There is another Python-based system called **DeepNeuralNetQSAR** that uses computational methods to help discover the molecular activity of chemicals. You can find more information about it at reference [117].

However, there is a program called Chemputer that assists in documenting the process of chemical synthesis in a consistent manner [118].

The **Tox21** Data Challenge assessed the harmful effects of 12,000 substances through the utilization of specifically engineered tests. Deep learning (DL) was employed in this example to forecast the likelihood of toxicity. The **DeepTox** software demonstrated improved performance compared to previous methods, providing evidence that deep learning is a superior approach for predicting toxicity [119].

AlphaFold, an artificial intelligence (AI) algorithm created by *DeepMind, a subsidiary of Alphabet*, is designed to forecast the three-dimensional (3D) configurations of proteins. The algorithm employs advanced deep learning techniques and has achieved remarkable success in predicting protein structures. AlphaFold competed in the Critical Assessment of Structure Prediction (CASP) competition and shown exceptional precision in forecasting protein structures, surpassing other participating teams [120]. AlphaFold has undergone two significant iterations: AlphaFold 1, released in 2018, and AlphaFold 2, released in 2020. Both versions of the algorithm exhibited outstanding performance in accurately predicting protein structures. AlphaFold 2, specifically, attained a significantly superior level of precision compared to all other participants in the CASP competition. For over two-thirds of the proteins in the global distance test (GDT), the result was higher than 90. The GDT measures how well the predicted structure corresponds with the empirically determined structure.

The achievement of AlphaFold is widely recognized as a major advancement in the field of structural bioinformatics. Nevertheless, it is crucial to acknowledge that despite AlphaFold's impressive precision, there are still obstacles to be tackled, and the issue of protein folding is not yet deemed fully resolved. However, the impressive technological accomplishment of AlphaFold has gained global admiration and has the capacity to expedite the progress of structural bioinformatics.

AlphaFold employs an AI-driven technique known as **Evoformer**, which considers protein structure prediction as a graph inference issue in three-dimensional space. This strategy utilizes modernized procedures that are sequentially applied within each block of the network. The pair representation in AlphaFold encapsulates data regarding the interconnections among the residues, whereas the MSA (multiple sequence alignment) model encapsulates data regarding individual residues and the sequences in which they are present. The MSA representation is updated by performing an element-wise outer product, which enables continuous communication between the changing MSA representation and the pair representation. This methodology allows for the precise forecasting of protein structures [121]. AlphaFold has the capacity to transform structural bioinformatics and expedite our comprehension of protein structures. By integrating artificial intelligence methodologies with well curated structure and sequence databases, it has the potential to narrow the divide between genomics and experimental structure determination. This can result in a more profound comprehension of biological processes and the advancement of novel therapeutic approaches [121].

ORGANIC [122] is a software program used for generating molecules that possess specific desired features at the molecular level. It employs generative machine learning techniques and deep neural networks to facilitate the creation of innovative molecular compounds. The utilization of deep neural networks and high-throughput simulations in a goal-directed generative model facilitates a focused design approach, reducing design bias and enabling the study of distinct material property spaces [123]. ORGANIC is mostly utilized in the development of organic electronic materials, namely for applications like organic light-emitting diodes (OLEDs). ORGANIC utilizes generative machine learning frameworks, including recurrent neural networks (RNNs) and deep reinforcement learning (RL), to quickly discover novel materials suitable for OLED applications. This methodology expedites the advancement of innovative organic electronic materials, while also facilitating extension into diverse fields such as catalyst design, aerospace, life science, and petrochemicals [123].

ORGANIC can also be utilized in the virtual design of organic semiconductors using metal-organic frameworks (MOFs). MOFs provide a method to control the arrangement of organic semiconductor molecules at a supramolecular level, enabling the customization of material properties. ORGANIC has the capability to streamline the design process through the generation of MOF structures, analysis of their structural dynamics, and prediction of attributes like as conduction, absorption, and light interaction. This automated workflow tool improves the modeling and characterization of a diverse range of MOFs [124].

ORGANIC's development offers researchers a potent tool for molecule creation and property optimization. By utilizing machine learning techniques and conducting large-scale simulations, it allows for the efficient investigation of chemical space and the creation of molecules with specific features. The usefulness and application of the software is further enhanced by its integration with current computational chemistry software, such as VASP and Gaussian, and the availability of user-friendly interfaces. This makes it suitable for usage in numerous research disciplines [125].

PotentialNet [126] is a software application that employs neural networks to forecast the strength of the interaction between ligands and protein–ligand complexes. The use of deep attention techniques and descriptor embeddings enhances the precision of binding affinity prediction [127]. The algorithm specifically targets intermolecular interactions within the molecular complex in order to acquire significant characteristics for precise prediction [127]. The methodology employed in PotentialNet entails encoding the chemical complex as a one-dimensional vector derived from the descriptor information acquired from the training dataset. Descriptors are generated for the protein and ligand atom pairs that are in interaction inside the complex. The Euclidean distance is used to calculate the distance between the protein atom and the ligand atom pair. Typically, a cutoff value of 12 Å is selected for this calculation [127]. The program takes into account different intermolecular interactions, such as steric interactions, hydrophobic interactions, and hydrogen bonding.

PotentialNet's model design includes an attention method that identifies significant descriptors for predicting binding affinity [128]. This method enhances the ability to identify and capture the specific locations where the protein and ligand interact, resulting in improved accuracy and effectiveness in predicting outcomes. The application employs a deep neural network model known as PUPResNet, which is composed of encoder and decoder blocks with skip connections [128].

In order to assess the effectiveness of the binding affinity models, many metrics are employed, such as the mean absolute error (MAE), root mean square error (RMSE), Pearson's correlation coefficient (PCC), Spearman's correlation coefficient (SCC), and the standard deviation in

regression (SD) [127]. The process of determining the most suitable number of descriptors is carried out by training a random forest model and arranging the descriptors based on their priority. The model is subsequently trained using several quantities of descriptors, and its performance is assessed to identify the most effective number [127].

Hit Dexter is a machine learning technique employed to forecast compounds that have the potential to react to biochemical assays. This is a machine learning model that is particularly created to identify potential hits throughout the drug discovery process. Hit discovery entails the identification of tiny molecules with the capability to attach to a certain target and modify its functionality. Hit Dexter utilizes a variety of algorithms and multiple criteria to detect hits on specific biological targets or neurological problems [35].

The employment of deep learning techniques in Hit Dexter is crucial for extracting and analyzing the chemical and physical properties of compounds. Deep learning models have been employed to forecast features, such as absorption, distribution, metabolism, and excretion (ADME), of potential medication candidates. These models integrate molecular graphs and employ ideas like Convolutional Neural Networks (CNNs) and Artificial Neural Networks (ANNs) to forecast features such as oral medication absorption. Accurate predictions can be made by utilizing different chemical descriptors and employing machine learning methods such as SVM regression and boosting [35].

Another feature of Hit Dexter is the utilization of Quantitative Structure-Activity Relationship (QSAR) models. QSAR models utilize deep learning approaches to extract features from chemical strings and automatically derive pertinent molecular descriptors. These models have demonstrated encouraging outcomes in hit-to-lead optimization studies, offering forecasts for binding affinities and assisting in the identification of new drug candidates [35].

The scope of Hit Dexter's application extends beyond drug discovery. Additionally, it can be applied in the forecasting of bioactivity, prognosis of prote-protein interactions, homology modeling/prediction of protein folding, and digital pathology [35].

DeltaVina [129] is a scoring function employed to reevaluate the binding affinity between proteins and ligands. The system utilizes machine learning approaches and a variety of algorithms to enhance the precision of forecasts regarding binding affinity [130]. DeltaVina is specifically engineered to enhance the precision of evaluating the protein-ligand binding affinity. The system employs a range of machine learning methods, such as XGBoost and random forest, to integrate diverse chemical descriptors and data. DeltaVina's performance has been assessed on benchmark datasets and compared to traditional scoring systems, showcasing its efficacy in forecasting binding affinities and finding promising pharmaceutical candidates. The scoring function and its code are accessible for utilization and additional investigation [131].

The neural graph fingerprint [132] is a method employed to forecast the characteristics of unfamiliar molecules. The process entails creating molecular sentences and encoding them using a language model, such as BERT, to get high-dimensional embeddings of the substructures. The embeddings depict the molecules as vectors, which can be utilized for subsequent tasks such as predicting chemical properties [133].

Once the molecular phrases are formed, they are encoded using a language model, such as BERT. This pre-trained model is capable of producing high-dimensional embeddings of the substructures, which represent the molecules in vector format. The embeddings effectively capture crucial

characteristics and structural details of the molecules, enabling accurate prediction of their diverse chemical properties [133]. To evaluate the performance of the prediction model, multiple measures are used. For example, for classification tasks, the cross-entropy loss function is utilized, and the prediction performance is evaluated using the ROC-AUC (the area under the receiver operating characteristic curve) [133]. The ROC-AUC quantifies the model's capacity to differentiate between positive and negative samples. On the other hand, for regression tasks, the mean squared error (MSE) loss function is utilized, and the prediction performance is assessed using the root mean squared error (RMSE) and the coefficient of determination (R²) [133]. The RMSE evaluates the average squared difference between the predicted and real property values, whereas the R² reflects the goodness of fit and prediction. The neural graph fingerprint, with its molecular sentence production and language model-based encoding, offers a potent tool for predicting the features of new compounds. By combining deep learning algorithms and molecular embeddings, it enables accurate predictions and the investigation of numerous chemical properties [133].

GastroPlus [134] is a platform that utilizes AI and predictive modeling to aid in the design and optimization of drug formulations. This software is used for pharmaceutical products (dosage form) in many animal models, including rats, primates, and animals, allowing researchers to predict the drug absorption, distribution, metabolism, and excretion [135]. GastroPlus's AI-driven approach aids in the development of more effective and bioavailable drug formulations, streamlining the drug development process.

3.2. Example of AI in Drug Design

Next, we will present some examples which cover a range of topics related to AI in drug design, including the integration of AI in drug discovery, the prediction of drug–protein interactions, and practical guidelines for using ML algorithms. They provide valuable insights into the use of AI in the pharmaceutical industry and offer potential solutions to challenges in drug design and development.

For example, Paul D. et al. discusses the integration of artificial intelligence in drug discovery and development, including the tools and techniques used. They also highlight ongoing challenges and ways to overcome them [14].

Dara S. et al. explores various applications of artificial intelligence in drug design, including in the prediction of drug–protein interactions, the discovery of drug efficacy, and ensuring safety biomarkers [35].

Jiménez-Luna et al. focuses on explaining compound activity predictions using a substructure-aware loss for graphing neural networks. They explore the use of deep learning in predicting molecular properties [24]. They also provide practical guidelines for using gradient boosting in molecular property prediction. They offer insights into the application of ML algorithms in drug design.

Schneider et al. discuss the foundation of artificial intelligence in therapeutic science, including its applications in drug design. They explore adaptive graph learning methods for automated molecular interactions and property predictions [136].

The iterative process to design the 3D structures of receptors to generate a novel molecule is termed *de novo* drug designing, which is intended to produce new dynamics. Researchers utilized the indolent space portrayal to prepare a model dependent on the quantitative estimate of the drug

likeness and drug similarity scores, and the manufactured availability score and synthetic accessibility score [137].

3.3. Examples Cases of AI Used in Polypharmacology

Next, we present some examples of papers that shed light on the application of AI in polypharmacology and drug discovery. They discuss how AI can be used to predict drug–protein interactions and design safer drug molecules. Understanding polypharmacology is crucial in developing drugs that are effective against multiple targets while minimizing off-target effects. These papers provide valuable insights into the advancements and challenges in this field and offer potential strategies for improving drug design and development. Chauhari et al. explain the concept of polypharmacology, which refers to the tendency of a drug molecule to interact with multiple receptors, leading to off-target effects. They explore how artificial intelligence can aid in the design of safer drug molecules by considering polypharmacology [138].

Reddy et al. [139] provide an overview of polypharmacology studies and the challenges involved. They discuss the potential of a rational design using polypharmacology to develop more effective and less toxic therapeutic agents. The review outlines the latest progress in this field. Other authors [140] explore polypharmacy using AI and data analysis. The AI axis focuses on defining and detecting polypharmacy using algorithms to predict health outcomes based on medication combinations.

The use of AI in pharmaceutical product development is also discussed in a review [14]. The review suggests that AI can replace the traditional trial-and-error method in designing formulations. Computational tools, including quantitative structure–relationship (QSPR) models and decision-support tools, can help address issues related to formulation design such as stability, dissolution, and porosity. Furthermore, the integration of AI with mathematical models like computational fluid dynamics (CFD) can speed up the production of pharmaceutical products [14].

The involvement of AI in the development of a pharmaceutical product from the bench to the bedside can be imagined given that it can aid rational drug design [141]; assist in decision making; determine the right therapy for a patient, including personalized medicines; and manage the clinical data generated and use it for future drug development [142]. E-VAI is an analytical and decision-making AI platform, developed by Eularis, which uses ML algorithms along with an easy-to-use user interface to create analytical roadmaps based on the competitors, key stakeholders, and currently held market share to predict key drivers in the sales of pharmaceuticals [143], thus helping marketing executives allocate resources for the maximum market share gain, reversing poor sales and enabling them to anticipate where to make investments.

3.4. Example Cases of AI in Drug Chemical Synthesis

The following example cases of AI in drug chemical synthesis highlight the use of artificial intelligence and ML in drug chemical synthesis. They explore the application of DL models in de novo drug design, the prediction of reaction outcomes, and the optimization of synthetic routes. The integration of AI with chemical knowledge and data-driven approaches has the potential to accelerate the drug discovery process and facilitate the design of new drug molecules. The paper by Paul D. et al. [14] is focused on the application of DL in drug discovery and medicinal chemistry, including the synthesis of new drug molecules. It explores the use of generative models and RL for de novo drug design and optimization.

Jiménez-Luna et al. [24] describe the use of AI in synthetic organic chemistry, specifically in reaction prediction, retrosynthesis, and reaction optimization. They discuss the integration of ML algorithms with chemical knowledge for efficient and automated synthesis planning. Their paper introduces DeepChem, a deep learning toolkit designed for drug discovery tasks. It covers various applications, including the generation of novel chemical structures, predicting chemical properties, and optimizing synthetic routes. The paper highlights the potential of deep learning in accelerating the drug discovery process. It also presents ML models for predicting the outcomes of organic reactions, including the yield and regioselectivity. It discusses the use of reaction fingerprints and molecular descriptors as input features for predicting reaction outcomes. The study demonstrates the potential of ML in guiding synthetic chemistry efforts.

3.5. Case Examples of AI Used in Drug Screening

These examples provide insights into the application of AI in drug screening and discovery. They highlight the use of AI algorithms for identifying novel compounds, designing new drug candidates, and predicting protein structures. The integration of AI in drug screening processes can help accelerate the discovery of potential drug candidates and optimize the drug discovery pipeline.

Paul D et al. [14] discuss the integration of AI in various sectors of the pharmaceutical industry, including drug discovery and development. They highlight the use of AI in drug screening and repurposing, reducing the human workload, and accelerating the achievement of targets. Their paper also discusses the ongoing challenges and the potential ways to overcome them, along with the future of AI in the pharmaceutical industry.

Blanco-González et al. [3] present case studies demonstrating the successful application of AI in drug discovery. They showcase examples where AI algorithms have been used to identify novel compounds for cancer treatment, inhibitors for specific proteins involved in diseases like Alzheimer's, and potential drugs for combating COVID-19. Their paper highlights the ability of AI to accelerate the drug discovery process and enable the development of more effective medications. It explores the impact of AI on the drug discovery process, with a focus on the design of novel compounds. It discusses how AI-based approaches, such as deep learning algorithms, can rapidly and efficiently design new drug candidates with desirable properties and activities. The paper also mentions the use of AI in structural biology, such as the development of AlphaFold, a software platform that uses AI to predict protein structures. This advancement has the potential to revolutionize personalized medicine and drug discovery.

Discovering drugs through AI models is a topic discussed on nexocode.com [144]. By analyzing large amounts of data, AI models can detect correlations that were previously unknown. This technology is being utilized in drug discovery to identify new drugs, design drug candidates, and assist in preclinical testing. AI is also being used to screen chemical repositories, identify new drug targets, and aid in computer-assisted molecular design.

A recent study showcased the use of AI in material discovery, as seen on unibas.ch [145]. The study's author, a doctoral student, utilized quantum mechanics to generate predictions for thousands of crystals that had randomly determined chemical compositions. These predictions served as training data for statistical ML models. The ML models demonstrated an accuracy level equivalent to standard quantum mechanical approaches, but with the added benefit of speed. In a day, ML

models can provide predictions, while quantum mechanical calculations would take a supercomputer over 20 million hours.

An interesting example is found on popularmechanics.com [146], where researchers used AI to develop a new type of compressible material. They did not have to rely on trial-and-error experimentation because they used existing research and AI algorithms to create new designs. This approach made the material discovery process much faster and allowed scientists to explore more design possibilities. The research also showed how important data-driven methods are in materials science. AI can suggest new approaches and designs as long as there are enough accurate data available.

The use of AI in the field of medicine goes beyond discovering new materials. According to ibm.com [147], AI has the potential to support precision medicine by providing patients with real-time recommendations that are tailored to their medical history, preferences, and needs. AI-powered virtual assistants are available 24/7 to provide healthcare information and address patient queries. Furthermore, AI is employed in medical imaging to detect diseases like breast cancer with comparable accuracy to human radiologists. Additionally, AI can help clinicians make sense of vast numbers of medical images and data, making them more manageable and enhancing patient care.

Healthcare is an ever-evolving field that is exploring the potential of AI in various medical domains. A comprehensive overview of AI in healthcare can be found on en.wikipedia.org [148]. In oncology, AI is used for cancer diagnosis, risk stratification, the molecular characterization of tumors, and drug discovery. For instance, AI algorithms have shown high accuracy in identifying breast cancer and prostate cancer. AI techniques such as artificial neural networks and Bayesian networks are used to diagnose and classify diseases. With the assistance of electronic health records, AI can analyze large amounts of data and help doctors with a patient's diagnosis and treatment. Physics-inspired ML approaches are also being explored to enhance medical diagnostic approaches and perform biomarker analysis.

To benchmark and test AI applications in the health domain, the United Nations (WHO/ITU) has established the ITU-WHO Focus Group on Artificial Intelligence for Health (FG-AI4H). They are working on various cases for its use such as in assessing breast cancer risk, guiding antivenom selection, and diagnosing skin lesions.

The market potential of AI applications in drug discovery and development is immense. AI has emerged as a powerful tool in the pharmaceutical industry, revolutionizing the way new drugs are discovered, developed, and brought to market. With its ability to analyze vast amounts of data, AI offers the potential to accelerate the drug development process, reduce costs, and improve the success rate of bringing new medicines to patients.

Pharmaceutical companies, research institutions, and startups are increasingly investing in AI technologies to streamline drug discovery processes and gain a competitive edge in the market. Partnerships between AI companies and pharmaceutical giants are becoming more common, fostering the development of innovative drug discovery platforms.

3.6. AI in Drug Discovery and Repurposing

In this section we provide information on different drugs discovered by means of AI, highlighting AI's significant role in drug discovery, accelerating the identification of potential drug candidates and optimizing drug development processes. In the following, we present some drugs that have been discovered or aided by AI.

Zong et al. [149] explore a computational drug repurposing application by using electronic health records (EHRs) to discover new applications for approved or investigational drugs. Their review identified four themes: publication venues, data types and sources, data processing and prediction methods, and targeted diseases, validation, and released tools. The utilization of EHRs in drug repurposing is hindered by validation, accessibility, and understanding issues.

Yadi et al. [150] introduce guidelines on how to use AI for accelerating drug repurposing or repositioning. They present how to use AI models in precision medicine, and as an example, how AI models can accelerate COVID-19 drug repurposing.

Wang et al. [151], a novel scoring algorithm, DrugRepo, uses chemical and genomic data to repurpose drugs for various diseases. In total, 516 approved drugs have been repurposed for 545 diseases using DrugRepo. Hundreds of novel predicted compounds have the potential for matching ongoing clinical trials.

Different research groups used AI algorithms to discover that naproxen, a commonly used pain reliever, has potential antiviral properties. An excellent illustration of an in silico and target-based technique for finding novel antivirals is naproxen. Lejal et al. used the X-ray structure of the RNA-free NP of H1N1 as a prototype to develop a structural-based modeling technique to find potential medications that target the nucleoprotein (NP) of the influenza A virus [152]. The non-steroidal anti-inflammatory medication naproxen, a well-known inhibitor of inducible cyclooxygenase type 2 (COX-2), was discovered using an in silico screening that was concentrated on a selected particular location of the NP structure. AI models identified this drug as potential inhibitors of a viral enzyme, providing valuable insights for future antiviral drug development [152]. In another study, naproxen demonstrated antiviral properties against SARS-CoV-2 using an in silico molecular docking analysis (using Autodock Vina) [153].

By using molecular modeling and virtual screening studies, Mostafa et al. tested the antiviral activity of anti-microbial and anti-inflammatory FDA-approved drugs, commonly prescribed to relieve respiratory symptoms, against SARS-CoV-2 [154]. The docking study was performed using OpenEye scientific software version 2.2.5. A virtual screening study illustrated that Azithromycin, Niclosamide, and Nitazoxanide bind to the main protease of SARS-CoV-2.

Baricitinib is an oral Janus kinase (JAK)1/JAK2 inhibitor approved for the treatment of rheumatoid arthritis. The Eli Lilly and Incyte Corporation used AI algorithms to analyze large-scale datasets to predict its usefulness on COVID-19 infection through planned anti-cytokine effects and as an inhibitor of host cell viral propagation [155].

DSP-1181 was Developed by Sumitomo Dainippon Pharma and Exscientia. It is a drug designed using AI algorithms [156]. It became the first AI-designed drug to enter human clinical trials for the treatment of obsessive-compulsive disorder (OCD). DSP-1181 is a long-acting, potent serotonin 5-HT1A receptor agonist [156]. AI played a pivotal role in the drug's discovery by rapidly analyzing vast datasets, identifying potential drug candidates, and optimizing the molecular structures to enhance their therapeutic effects. The successful progression of DSP-1181 to clinical trials showcases the potential of AI in expediting drug development for challenging neurological disorders [156].

DSP-2230 was also Developed by Sumitomo Dainippon Pharma and Exscientia, by means of AI-design, and is being investigated for the treatment of OCD. AI algorithms were utilized to analyze

biological data and identify the novel compounds that can modulate the specific brain receptors implicated in the disorder [157].

Berzosertib (formerly M6620, VX-970), developed by Merck KGaA and C4X Discovery [158], is a promising drug investigated for its applications in targeting the DNA repair mechanisms in various cancers. AI played a significant role in identifying Berzosertib as a potential candidate for cancer treatment [159]. Through the AI-driven analysis of genetic and molecular data, researchers could pinpoint specific DNA repair pathways that the drug could effectively target, making it a promising addition to cancer therapy.

Halicin is a revolutionary antibiotic discovered by MIT researchers through the application of AI models [160]. It demonstrated remarkable effectiveness against a wide range of bacterial infections, including drug-resistant strains [161]. AI enabled the screening of a vast number of chemical compounds to identify Halicin as a potent antibiotic. The discovery of Halicin highlights the potential of AI in quickly identifying new classes of antibiotics to combat the growing problem of antibiotic resistance.

Evinacumab is a drug developed by Regeneron Pharmaceuticals using AI [162]. The drug was identified as a potential therapy for homozygous familial hypercholesterolemia (a rare genetic disorder characterized by extremely high cholesterol levels), refractory hypercholesterolemia (both familial and non-familial), and severe hypertriglyceridaemia [162]. AI algorithms were utilized to analyze vast datasets of genetic and biochemical information to identify the specific targets and pathways involved in cholesterol regulation. This helped researchers identify evinacumab as a monoclonal antibody that binds to and blocks an enzyme called ANGPTL3, thereby reducing cholesterol levels in affected individuals [162].

The Amarin Corporation developed **Icosapent Ethyl** to reduce cardiovascular risk [163]. The AI-driven analysis of clinical trial data helped demonstrate the drug's effectiveness in reducing cardiovascular events in patients with high triglyceride levels, leading to its approval by regulatory authorities. Icosapent Ethyl is a high-purity prescription form of EPA ethyl ester approved by the US Food and Drug Administration at a dose of 4 g/day as an adjunct to diet to reduce plasma triglyceride levels in adults with severe hypertriglyceridemia [164].

LegoChem Biosciences, using AI models, developed **Delpazolid (LCB-010371)** as a potential antibiotic with effectiveness against drug-resistant bacteria, including Methicillin-resistant *Staphylococcus aureus* (MRSA) and Vancomycin-resistant Enterococci (VRE) [165]. AI-driven drug discovery methods were employed to identify Delpazolid as a novel antibiotic candidate. LegoChem Biosciences utilizes its ConjuAll and Legochemistry platform technologies to discover and develop drugs.

InSilico Medicine has used their AI platform **Pharma.AI™** to develop INS018-055 to help treat idiopathic pulmonary fibrosis [166]. Targeting the discoidin domain receptor 1 (DDR1) is how INS-08055 works. DDR1, an epithelial cell-expressed pro-inflammatory receptor tyrosine kinase involved in fibrosis, is activated by collagen and inhibited by INS-08055. The medication candidate, which is delivered orally and intravenously, improves illness symptoms by inhibiting DDR1 [166].

Developed by Merck & Co., **Pembrolizumab (Keytruda)** is an AI-discovered drug used for the treatment of various cancers, including melanoma, lung cancer, and head and neck cancers [167].

The UK Medicines and Healthcare Products Regulatory Agency (MHRA) has received a clinical trial application from *BenevolentAI* for **BEN-8744**, a small molecule phosphodiesterase 10 (PDE10) inhibitor that is intended to treat ulcerative colitis [168].

Sotorasib (AMG 510), developed by *Amgen*, is an AI-optimized drug used for the treatment of non-small cell lung cancer with specific genetic mutations. AI algorithms were used to analyze cancer genomic data and identify potential compounds that selectively inhibit the mutated protein responsible for tumor growth [169].

The above-mentioned drug examples highlight the diverse applications of AI in drug discovery and development, from optimizing drug structures and formulations to identifying novel targets and repurposing existing drugs for new indications. AI technologies continue to revolutionize the pharmaceutical industry, enabling more efficient and targeted drug development processes that hold the potential to transform patient care and treatment outcomes.

4. Literature Review

(**Adarsh Sahu et al., 2022**) studied Artificial Intelligence (AI) in Drugs and Pharmaceuticals, The AI is an effective tool for data mining based on the huge pharmacological data and machine learning process. Hence, AI has been used in de novo drug design, activity scoring, virtual screening and in silico evaluation in the properties (absorption, distribution, metabolism, excretion and toxicity) of a drug molecule. Various pharmaceutical companies have teamed up with AI companies for faster progress in the field of drug development, along with the healthcare system. The review covers various aspects of AI (Machine learning, Deep learning, Artificial neural networks) in drug design. It also provides a brief overview of the recent progress by the pharmaceutical companies in drug discovery by associating with different AI companies.

(**Shruti Singh et al., Aug 2023**) reviewed Artificial Intelligence and Machine Learning in Pharmacological Research: Bridging the Gap Between Data and Drug Discovery, The responsible deployment of AI necessitates robust frameworks and regulations. The future of AI in pharmacological research is promising, with integration with emerging technologies like genomics, proteomics, and metabolomics offering the potential for personalized medicine and targeted therapies. Collaboration among academia, industry, and regulatory bodies is essential for the ethical implementation of AI in drug discovery and development. Continuous research and development in AI techniques and comprehensive training programs will empower scientists and healthcare professionals to fully exploit AI's potential, leading to improved patient outcomes and innovative pharmacological interventions.

(**Molly Bekbolatova et al., Jan 2024**) reviewed Transformative Potential of AI in Healthcare: Definitions, Applications, and Navigating the Ethical Landscape and Public Perspectives, AI in healthcare should not be viewed as a threat to human workers. Instead, AI systems are designed to augment and support healthcare professionals, freeing up their time to focus on more complex and critical tasks. By automating routine and repetitive tasks, AI can alleviate the burden on healthcare professionals, allowing them to dedicate more attention to patient care and meaningful interactions. However, legal and ethical challenges must be addressed when embracing AI technology in medicine, alongside comprehensive public education to ensure widespread acceptance.

(**Chayna Sarkar et al., Jan 2023**) reviewed Artificial Intelligence and Machine Learning Technology Driven Modern Drug Discovery and Development, The discovery and advances of medicines may be considered as the ultimate relevant translational science effort that adds to human invulnerability and happiness. DL-facilitated methods have just initiated kickstarting for some

integral issues in drug discovery. Many technological advances, such as “message-passing paradigms”, “spatial-symmetry-preserving networks”, “hybrid de novo design”, and other ingenious ML exemplars, will definitely come to be pervasively widespread and help dissect many of the biggest, and most intriguing inquiries. Open data allocation and model augmentation will exert a decisive hold during the progress of drug discovery employing AI. This review will address the impending utilizations of AI to refine and bolster the drug discovery operation.

(Rohan Gupta et al., Aug 2021) reviewed Artificial intelligence to deep learning: machine intelligence approach for drug discovery, The artificial neural network, deep neural network, support vector machines, classification and regression, generative adversarial networks, symbolic learning, and meta-learning are examples of the algorithms applied to the drug design and discovery process. Artificial intelligence has been applied to different areas of drug design and development process, such as from peptide synthesis to molecule design, virtual screening to molecular docking, quantitative structure-activity relationship to drug repositioning, protein misfolding to protein-protein interactions, and molecular pathway identification to polypharmacology. Artificial intelligence principles have been applied to the classification of active and inactive, monitoring drug release, pre-clinical and clinical development, primary and secondary drug screening, biomarker development, pharmaceutical manufacturing, bioactivity identification and physiochemical properties, prediction of toxicity, and identification of mode of action.

(Tianhua Yao et al., Jan 2024) Studied Deep evolutionary fusion neural network: a new prediction standard for infectious disease incidence rates, The infectious disease dataset is from an official release and includes four national and three regional datasets. The Baidu index platform provides internet data. We choose a single model (seasonal autoregressive integrated moving average (SARIMA), nonlinear autoregressive neural network (NAR), and long short-term memory (LSTM)) and a deep evolutionary fusion neural network (DEFNN). The DEFNN is built using the idea of neural evolution and fusion, and the DEFNN + is built using multisource data. We compare the model accuracy on reference group data and validate the model generalizability on external data. (1) The loss of SA-LSTM in the reference group dataset is 0.4919, which is significantly better than that of other single models. (2) The loss values of SA-LSTM on the national and regional external datasets are 0.9666, 1.2437, 0.2472, 0.7239, 1.4026, and 0.6868. (3) When multisource indices are added to the national dataset, the loss of the DEFNN + increases to 0.4212, 0.8218, 1.0331, and 0.8575.

(Kriti Das et al., Dec 2023) reviewed Current implications and challenges of artificial intelligence technologies in therapeutic intervention of colorectal cancer, Artificial intelligence (AI) is an enormously in-demand, cost-effective technology, that uses various tools machine learning (ML), and deep learning (DL) to screen, diagnose, and stage, and has great potential to treat CRC. Moreover, different ML algorithms and neural networks [artificial neural network (ANN), k-nearest neighbors (KNN), and support vector machines (SVMs)] have been deployed to predict precise and personalized treatment options. This review examines and summarizes different ML and DL models used for therapeutic intervention in CRC cancer along with the gap and challenges for AI.

(Shao Jinsong et al., Feb 2024) reviewed Molecular fragmentation as a crucial step in the AI-based drug development pathway, This review provides a comprehensive overview of the current state of the art in molecular fragmentation. We systematically summarize the approaches and applications of various molecular fragmentation techniques, with special emphasis on the characteristics and scope of applicability of each technique, and discuss their applications. We also provide an outlook on the current development trends of molecular fragmentation techniques, including some potential research directions and challenges.

(**Xiaoying Yan et al., Jan 2024**) Discussed Predicting Drug-drug Interaction with Graph Mutual Interaction Attention Mechanism, Therefore, in this work, we propose a novel Graph learning framework of Mutual Interaction Attention mechanism (called GMIA) to predict DDIs by effectively representing the drug molecules. Specifically, we build the node-edge message communication encoder to aggregate atom node and the incoming edge information for atom node representation and design the mutual interaction attention decoder to capture the mutual interaction context between molecular graphs of drug pairs. GMIA can bridge the gap between two encoders for the single drug molecules by attention mechanism. We also design a co-attention matrix to analyze the significance of different-size substructures obtained from the encoder-decoder layer and provide interpretability. In comparison with other recent state-of-the-art methods, our GMIA achieves the best results in terms of area under the precision-recall-curve (AUPR), area under the ROC curve (AUC), and F1 score on two different scale datasets. The case study indicates that our GMIA can detect the key substructure for potential DDIs, demonstrating the enhanced performance and interpretation ability of GMIA.

(**Cigdem Alka et al., Jan 2024**) Computer-Selected Antiviral Compounds: Assessing In Vitro Efficacies against Rift Valley Fever Virus, As anticipated, these compounds did not impede viral RNA replication when administered three hours after infection. However, significant inhibition of viral RNA replication occurred upon viral entry when cells were pretreated with these small molecules. Furthermore, these compounds exhibited significant inhibition against Arumowot virus, another phlebovirus, while showing no antiviral effects on tick-borne bandaviruses. Our study validates AI-based virtual high throughput screening as a rational approach for identifying effective antiviral candidates for Rift Valley fever virus and other bunyaviruses.

(**Sarfaraz K Niazi and Zamara Mariam, Dec 2023**) reviewed Computer-Aided Drug Design and Drug Discovery: A Prospective Analysis, Emerging technologies like quantum computing, immersive technologies, and green chemistry promise to redefine the future of CADD. The trajectory of CADD, marked by rapid advancements, anticipates challenges in ensuring accuracy, addressing biases in AI, and incorporating sustainability metrics. This paper concludes by highlighting the need for proactive measures in navigating the ethical, technological, and educational frontiers of CADD to shape a healthier, brighter future in drug discovery.

(**Marwa H S Dawoud et al., Aug 2023**) Formulated Integrating Artificial Intelligence with Quality by Design in the Formulation of Lecithin/Chitosan Nanoparticles of a Poorly Water-Soluble Drug, The optimized formula was tested for its cytotoxicity. A design space was established, with an optimized formula having a molar ratio of 18.33:1 lecithin:chitosan and 38.35 mg silymarin. This resulted in nanoparticles with a size of 161 nm, a polydispersity index of 0.2, and an entrapment efficiency of 97%. The optimized formula showed a zeta potential of +38 mV, with well-developed spherical particles. AI successfully showed high prediction ability of the drug's release rate. The optimized formula showed an enhancement in the cytotoxic effect of silymarin with a decreased IC50 compared to standard silymarin. Lecithin/chitosan nanoparticles were successfully formulated, with deep process and product understanding. Several tools were used as AI which could shift pharmaceutical formulations from experience-dependent studies to data-driven methodologies in the future.

(**Lalitkumar K Vora et al., July 2023**) reviewed Artificial Intelligence in Pharmaceutical Technology and Drug Delivery Design, This review provides an overview of various AI-based approaches utilized in pharmaceutical technology, highlighting their benefits and drawbacks. Nevertheless, the continued investment in and exploration of AI in the pharmaceutical industry offer exciting prospects for enhancing drug development processes and patient care.

(**Muhammad Ahmer Raza et al., Dec 2022**) reviewed Artificial Intelligence (AI) in Pharmacy: An Overview of Innovations, AI has revolutionized healthcare to be more effective and efficient and the pharmacy sector is not left out. During the past few years, a considerable amount of increasing interest in the uses of AI technology has been identified for analyzing as well as interpreting some important fields of pharmacy like drug discovery, dosage form designing, polypharmacology, and hospital pharmacy. Given the growing importance of AI, we wanted to create a comprehensive report which helps every practicing pharmacist understand the biggest breakthroughs which are assisted by the deployment of this field.

(**Latifa Douali et al., Jan 2024**) studied Machine Learning to Predict Teratogenicity: Theory and Practice, It is sought as robust boosting methods to overcome many limitations and restrictions facing the experimental studies. By performing tasks such as classification, regression, clustering, anomaly detection, and decision systems, ML can be used to assess whether an agent is teratogen or not or to determine its teratogenic potential. It may also be used for the purpose of deciding on the use of medicinal products. In this chapter, we describe how ML can be used to investigate teratogenicity.

(**Prachi P Parvatikar et al., Dec 2023**) reviewed Artificial intelligence: Machine learning approach for screening large database and drug discovery, Although there are presently a wide variety of data-driven AI/ML tools available, the majority of these tools have, up to this point, been developed in the context of non-communicable diseases like cancer, and a number of obstacles have prevented the translation of these tools to the discovery of treatments against infectious diseases. In this review various aspects of AI and ML in virtual screening of large databases were discussed. Here, with an emphasis on antivirals as well as other disease, offers a perspective on the advantages, drawbacks, and hazards of AI/ML techniques in the search for innovative treatments.

(**Qi Lv et al., Dec 2023**) reviewed Artificial intelligence in small molecule drug discovery from 2018 to 2023: Does it really work?, This review aims to present a comprehensive overview of the utilization of AI methods in the field of small drug design, with a specific focus on four key areas: protein structure prediction, molecular virtual screening, molecular design, and absorption, distribution, metabolism, excretion, and toxicity (ADMET) prediction. Additionally, the role and limitations of AI in drug development are explored, and the impact of AI on decision-making processes is studied. It is important to note that while AI can bring numerous benefits to the early stage of drug development, the direction and quality of decision-making should still be emphasized, as AI should be considered as a tool rather than a decisive factor.

5. PharmaPredict_AI A Machine Learning Platform for Assessing Extensive Chemical Libraries

Drug discovery is extremely challenging. Identifying a molecule that can treat a disease is already a difficult task, but for a drug to be successful, it must do more than just treat the disease. The molecule needs to be absorbed by the body, reach the diseased area, exert its intended effect, and then be eliminated from the body, all without causing significant harm. Testing all these essential drug-like properties in the lab is a slow and costly process, so being able to predict them computationally would be incredibly advantageous.

We developed PharmaPredict AI to address this challenge. PharmaPredict AI is a machine learning platform designed to predict the absorption, distribution, metabolism, excretion, and toxicity

(ADMET) properties of a molecule, which are crucial for a drug to be both safe and effective. To ensure PharmaPredict AI is accessible to the research community, we created a free website. We encourage you to try it out and continue reading for more details!

5.1. Developing PharmaPredict AI

In the past, we have successfully used graph neural networks (GNNs) to predict a variety of molecular properties, from quantum mechanical energies to human toxicity. For PharmaPredict AI, we chose to use the same model architecture, known as Chemprop-RDKit, to predict ADMET properties.

5.2. How Chemprop-RDKit works

Chemprop-RDKit (Figure 6) incorporates a GNN called Chemprop, which computes basic features for each atom (such as atom type) and each bond (such as bond type). It then performs several steps of message passing with neural network layers to aggregate atom and bond information throughout the molecule to create a single representation of the entire molecule. This representation is then combined with 200 molecular features computed by RDKit, and the combined representation is processed through several feed-forward neural network layers to predict molecular properties. This hybrid architecture allows Chemprop-RDKit to leverage the strengths of GNNs (understanding local molecular environments) and RDKit (calculating features that describe the whole molecule).

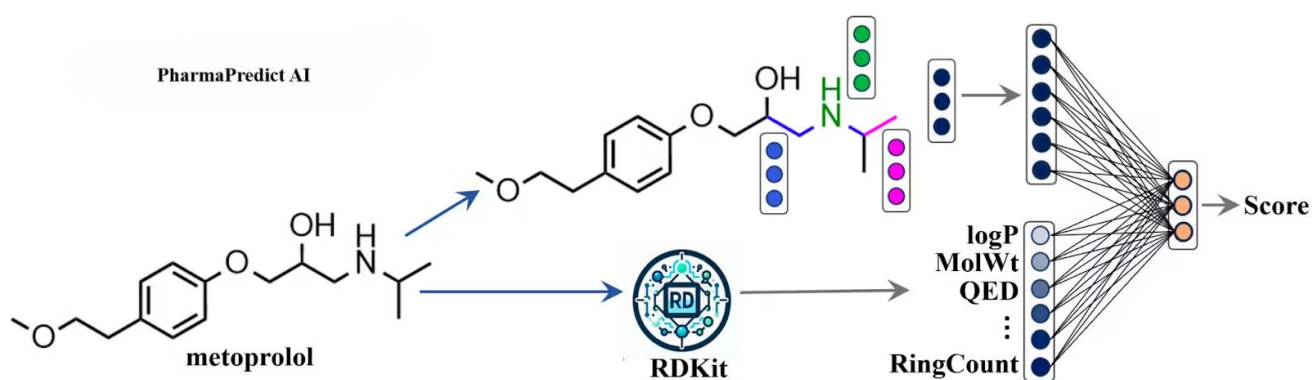


Figure 6. PharmaPredict AI utilizes machine learning models based on the Chemprop-RDKit architecture. Chemprop-RDKit integrates a graph neural network (top) with 200 physicochemical properties calculated by RDKit (bottom), which are then combined through a feed-forward neural network (right) to predict the properties of a molecule.

5.3. Training on Benchmark ADMET Data

We trained PharmaPredict AI's Chemprop-RDKit models using ADMET data from the Therapeutics Data Commons (TDC), a valuable resource of publicly available datasets for drug discovery. The TDC features a leaderboard called the ADMET Benchmark Group, which includes 22 ADMET datasets and is ideal for comparing different machine learning models for ADMET prediction.

To create the most comprehensive ADMET prediction model possible, we also incorporated an additional 19 ADMET datasets from the TDC that are not part of the leaderboard, resulting in a total of 41 ADMET datasets (10 regression and 31 classification). Additionally, we constructed two multi-task ADMET datasets: one comprising all 10 regression datasets and another comprising all

31 classification datasets. This approach allowed us to train multi-task models, which are often faster and more accurate than training numerous single-task models

5.4. Building the PharmaPredict AI Website

To ensure fast and accurate ADMET prediction is accessible to researchers, regardless of their computational expertise, we developed the user-friendly PharmaPredict AI website. On the website, you can input up to 1,000 molecules at a time by entering SMILES, uploading a CSV file, or using an interactive tool to draw molecules (Figure 3). You can then generate predictions and either view them on the website or download them as a CSV. The website features a radar plot summary of five key drug-like properties (Figure 4) and provides prediction values for all 41 ADMET properties (Figure 5).

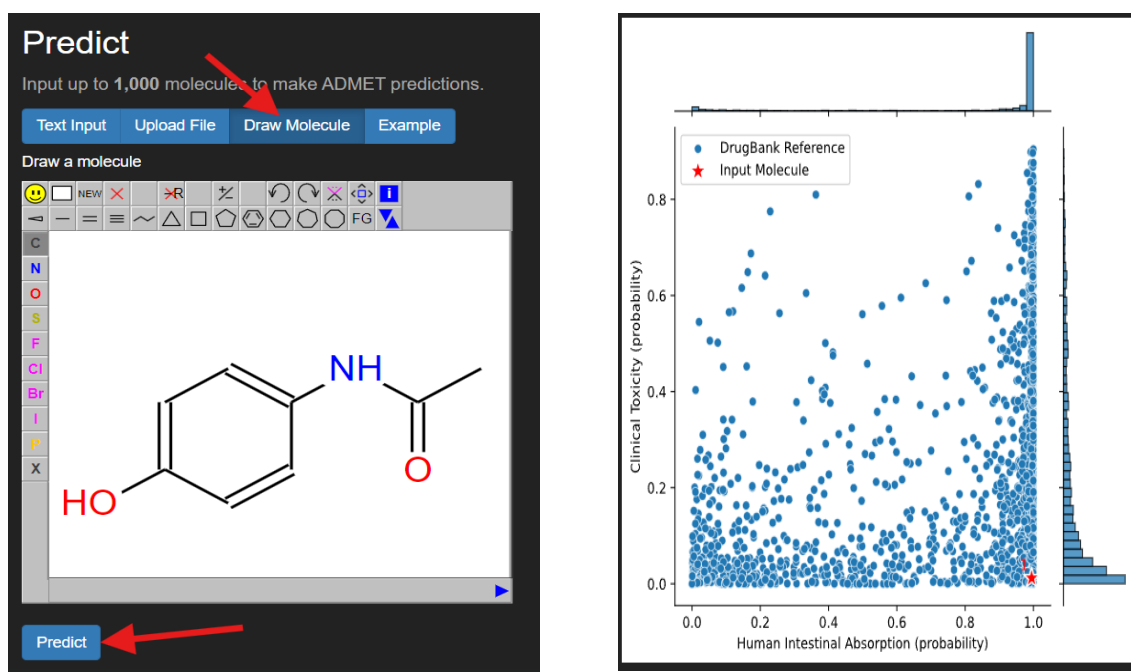


Figure 7. On the PharmaPredict AI website, users can input up to 1,000 molecules at a time by entering SMILES, uploading a CSV file, or drawing molecules using an interactive tool.

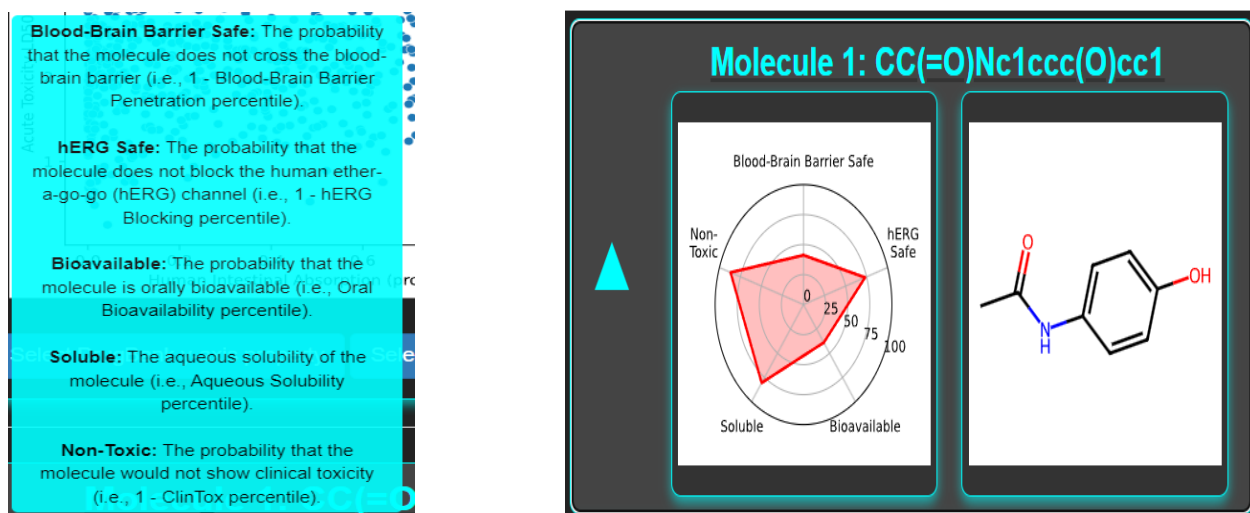


Figure 8. PharmaPredict AI displays a radar plot summarizing five key drug-like properties of the molecule along with a visualization of the molecule's structure



Figure 9. PharmaPredict AI displays computed values for eight physicochemical endpoints along with predicted values for 41 ADMET properties (shown in the “Value” column). PharmaPredict AI also calculates the percentile of each prediction compared to 2,579 approved drugs in the DrugBank (listed in the “DrugBank Percentile” column). The percentile offers valuable context for understanding how the molecule compares to approved drugs. (Note: Some ADMET properties are not displayed in this figure to save space.)

5.5. Contextualizing ADMET Predictions

Interpreting ADMET predictions can be challenging because different categories of drugs have varying requirements for their molecular properties. For instance, high toxicity may be acceptable for chemotherapies due to the lack of better alternatives, but it is generally unacceptable for antibiotics, as their toxicity can outweigh their benefits compared to safer options.

To provide context for PharmaPredict AI's predictions, we decided to compare predictions on new molecules to those on known drugs. We created a reference set of 2,579 drugs from DrugBank that have obtained regulatory approval and used PharmaPredict AI to make ADMET predictions for all these molecules. Now, when PharmaPredict AI generates predictions for a new molecule, it also calculates the percentile of these predictions relative to the 2,579 approved drugs (Figure 5). For example, a molecule with a toxicity percentile of 90% is more toxic than 90% of approved drugs.

To offer users even more detailed context, we added an option to select an Anatomical Therapeutic Chemical (ATC) code, such as “antibiotics for topical use,” to create a more specific reference set for computing percentiles (Figure 6). By selecting an ATC code, users can compare their proposed molecule to only the approved drugs labeled with that ATC code, allowing them to see if their molecule has similar ADMET properties to approved drugs for the same therapeutic use.

6. The Market Prospects of AI-Based Drug Development

Pharmaceutical companies are converting to AI solutions to reduce the financial costs and potential losses connected with virtual screening. Between 2015 and 2018, the AI market grew from USD 200 to USD 700 million, and by 2024, it is expected to reach USD 5 billion [170]. An expected 40% increase from 2017 to 2024 suggests that AI may reshape the pharmaceutical and medical industries. Many pharmaceutical companies have invested in AI or are continuing to do so, as well as collaborating with AI service providers to develop critical medical equipment. Examples of collaboration are given in.

7. Challenges and Limitations in AI-Assisted Drug Discovery

The use of AI to aid in the discovery of new materials holds immense potential in developing materials with specific properties for diverse applications. However, there are challenges and limitations that need to be acknowledged and addressed to ensure its effective implementation [3].

One of the main challenges highlighted in our paper is related to the quality and quantity of data. AI models require a large volume of high-quality data for training. In materials science, obtaining comprehensive and dependable datasets can be difficult due to the vast number of materials and properties that can be taken into consideration. Incomplete or inaccurate data can result in biased or erroneous predictions.

Therefore, reliable and high-quality data are the foundation of AI-assisted research. Industries must ensure that the data used for training AI models are precise, representative, and related to the specific research objectives. It is crucial to address data limitations, such as biases or incomplete datasets, to avoid skewed results and unreliable conclusions.

When it comes to discovering new materials, there are several important factors to consider. Firstly, data representation is key. This involves transforming complex material data, like crystal structures or chemical compositions, into formats that AI models can understand. It is important to develop representations that capture essential features while remaining interpretable [171].

Algorithm selection and development is also crucial. The appropriate AI algorithms for a material's discovery will depend on the specific problem at hand. Developing custom algorithms that can handle the unique challenges of material data, such as high dimensionality and non-linear relationships, requires expertise in both AI and materials science.

While AI models can predict material properties with impressive accuracy, interpreting the underlying reasons for these predictions remains a challenge. Understanding the relationship between the input features and output properties is essential for guiding further research and design.

Finally, effective material discovery often requires the integration of domain-specific knowledge, such as quantum mechanics or thermodynamics. Combining AI techniques with specialized scientific insights is essential to ensure accurate predictions and meaningful discoveries [14].

When training data are biased, AI models may not perform well when presented with new materials or conditions. It is crucial to ensure diversity and representativeness in the training dataset to avoid this issue. However, researchers may face limitations in accessing a powerful computing infrastructure, which is often necessary for resource-intensive tasks, such as simulating material properties at different scales or conducting high-throughput calculations. Additionally, the vast and intricate search space of potential materials can pose a challenge when trying to find the optimal candidate. Advanced optimization techniques are required to explore this space efficiently [172].

It is also important to validate AI model predictions through experimental testing, which can be a time-consuming and costly process. Moreover, discovering new materials with advanced properties raises ethical and safety concerns. AI-assisted discoveries must be safe, environmentally friendly, and compliant with regulations. Industries must establish ethical guidelines that ensure responsible data usage, transparent algorithms, and the fair treatment of all stakeholders to address these concerns. Overall, AI-assisted research presents exciting opportunities but also requires the careful consideration of these issues [173]. When incorporating AI tools into the material discovery process, researchers must adjust their workflows and thinking. Cooperation between materials scientists and AI specialists is crucial for the optimal usage of these tools. AI-assisted material discovery offers exciting possibilities for revolutionizing the field by speeding up the identification of materials with the desired properties. However, the challenges and limitations mentioned above stress the importance of interdisciplinary collaboration, better data organization, innovative algorithms, and ethical and scientific considerations. Overcoming these obstacles will lead to the successful integration of AI in material discovery and ultimately fuel innovation across various industries [174].

Industries that handle sensitive or proprietary data should prioritize data security and implement strong measures to safeguard information against unauthorized access and breaches. It is crucial to balance the benefits of AI-powered insights with data security concerns. To assure the reliability and applicability of AI-generated insights across multiple contexts, stringent validation techniques are necessary. Since AI-assisted research generally involves large computational resources, including sophisticated hardware and significant energy consumption, enterprises need to assess their computational needs and distribute resources appropriately. Industries should incorporate ways to identify and eliminate biases in AI models that may unwittingly perpetuate the prejudices existing in the training data. In regulated businesses, AI-assisted research must comply with appropriate norms and standards. To gain the full benefits of AI-driven research findings, industry must commit to long-term plans that demand continual investment in updating models, refining algorithms, and adjusting to new research needs. In promoting a collaborative atmosphere, enterprises must

encourage researchers to work alongside AI systems, utilizing their strengths to make better informed decisions and discoveries, and boosting human talents [3]. To sum up, applying AI in research can lead to new discoveries and drive innovation in numerous industries. However, it is vital to carefully evaluate important elements such as data quality, ethics, competence, security, and teamwork to enable successful implementation. By deliberately addressing these challenges, industries may fully use the potential of AI to alter their research procedures and reach unprecedented levels of discovery and innovation [3].

8.Future Directions and Opportunities in AI-Assisted Drugs Discovery

The global AI-in-drug discovery market is predicted to observe considerable expansion in the future years. Advancements in AI technologies, including as ML, natural language processing, and deep learning, have led to the development of sophisticated algorithms capable of assessing complicated biological data and predicting the interactions between medications and their targets.

AI-driven platforms are being utilized for the virtual screening of massive compound libraries, identifying prospective therapeutic candidates with more accuracy and efficiency than traditional methods. These platforms can also optimize lead compounds and anticipate their pharmacokinetic and toxicity characteristics, enabling in the selection of the most promising candidates for further development.

Furthermore, AI is changing drug design by enabling the development of unique molecular structures with specified features. AI-generated compounds can be tuned to target specific diseases and biological processes, bringing new prospects for precision medicine.

As regulatory authorities, such as the FDA, continue to embrace AI applications in drug development, the market potential of AI in the pharmaceutical business is likely to rise further. However, hurdles persist, including data privacy concerns, the validation of AI models, and the integration of AI with traditional drug development methods.

Moreover, AI's potential extends beyond medication research and development. It also plays a significant part in clinical trials, patient stratification, and the real-time monitoring of therapy reactions, enabling tailored medical methods that increase patient outcomes.

The application of advanced simulation and modeling techniques in AI has the potential to change the drug discovery process, making it more efficient and successful. With AI, it is able to model and simulate biological networks, which increases our understanding of disease mechanisms and aids in the discovery of new medications [175].

The application of AI in drug discovery offers several benefits. Traditional approaches confront constraints such as high prices, inefficiency, and lengthy timescales. AI can address these difficulties by automating and streamlining processes, resulting in more efficient and effective drug discovery. This facilitates the discovery of innovative medication candidates that may have been ignored using traditional methodologies. AI uses breakthroughs in biology and computers to produce cutting-edge algorithms for drug discovery, leveling the playing field and enabling more researchers to access sophisticated computational tools for drug development. Furthermore, AI models have a stronger predictive capability, minimizing the risk for false positives in drug screening. By carefully setting the test conditions, AI can detect meaningful interactions and rank potential candidates for further exploration [176].

The application of AI in drug development shows potential by transferring drug screening from the lab to a virtual environment, resulting in faster screening and the capacity to identify intriguing targets without substantial effort and labor. However, adopting advanced simulation and modeling approaches in AI for drug development faces obstacles such as the data quality, interpretability of AI models, and the requirement for validation and reproducibility. Despite these obstacles, AI has the ability to improve drug discovery speed, enable unbiased exploration, and enhance its predictive capacity, leading to decreased costs and an improved likelihood of successful drug discoveries. To successfully integrate AI in drug discovery, resolving these problems and assuring the dependability and interpretability of AI models is key.

9. Conclusions:

The future of AI in drug discovery requires closer integration with automation, enabling AI systems to make decisions on compound design and synthesis without human involvement. This shift from an augmented to an autonomous drug design paradigm has the potential to accelerate the drug discovery process and provide better starting points for drug development. Ultimately, the goal is to construct fully autonomous laboratories capable of iterating through the design–make–test–analyze cycle independently. This could lead to faster and more efficient drug discovery processes, with AI systems autonomously recommending and testing new molecules. However, challenges remain, such as ensuring the reliability and reproducibility of AI-assisted findings. Additionally, robust datasets and investments in AI technology are crucial for the future success of AI in drug research.

PharmaPredict AI exemplifies this integration by offering a machine learning platform that predicts the ADMET properties of molecules, critical for developing safe and effective drugs. By leveraging Chemprop-RDKit architecture, PharmaPredict AI combines the strengths of graph neural networks and physicochemical property computations to provide accurate predictions. The user-friendly website allows researchers to input up to 1,000 molecules and obtain comprehensive ADMET predictions, contextualized against a reference set of approved drugs from DrugBank. This context helps researchers understand how their molecules compare to existing drugs, enhancing decision-making in drug development.

Despite obstacles and the need for further work to fully integrate AI tools into the drug discovery cycle, AI has already shown promise in discovering new material combinations, accelerating candidate searches, optimizing formulations, improving target identification, and enhancing virtual screening. The future outlook for AI in drug discovery is optimistic, with breakthroughs anticipated in autonomous decision-making and the integration of AI with automation. PharmaPredict AI's capabilities demonstrate the potential of AI to revolutionize drug discovery, ultimately leading to more efficient and cost-effective processes.

10. References

1. Kaul, V.; Enslin, S.; Gross, S.A. History of artificial intelligence in medicine. *Gastrointest. Endosc.* **2020**, *92*, 807–812. [Google Scholar] [CrossRef]
2. Rudrapal, M.; Kirboga, K.K.; Abdalla, M.; Maji, S. Explainable artificial intelligence-assisted virtual screening and bioinformatics approaches for effective bioactivity prediction of phenolic cyclooxygenase-2 (COX-2) inhibitors using PubChem molecular fingerprints. *Mol. Divers.* **2024**. [Google Scholar] [CrossRef] [PubMed]
3. Blanco-González, A.; Cabezón, A.; Seco-González, A.; Conde-Torres, D.; Antelo-Riveiro, P.; Piñeiro, Á.; Garcia-Fandino, R. The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies. *Pharmaceuticals* **2023**, *16*, 891. [Google Scholar] [CrossRef]
4. Malik, P.; Pathania, M.; Rathaur, V.K. Overview of artificial intelligence in medicine. *J. Fam. Med. Prim. Care* **2019**, *8*, 2328–2331. [Google Scholar] [CrossRef]
5. Briganti, G.; Le Moine, O. Artificial Intelligence in Medicine: Today and Tomorrow. *Front. Med.* **2020**, *7*, 27. Available online: <https://www.frontiersin.org/articles/10.3389/fmed.2020.00027> (accessed on 20 October 2023). [CrossRef]
6. Yoon, H.J.; Jeong, Y.J.; Kang, H.; Jeong, J.E.; Kang, D.-Y. Medical Image Analysis Using Artificial Intelligence. *Prog. Med. Phys.* **2019**, *30*, 49–58. [Google Scholar] [CrossRef]
7. Wu, J.; Chen, J.; Cai, J. Application of Artificial Intelligence in Gastrointestinal Endoscopy. *J. Clin. Gastroenterol.* **2021**, *55*, 110–120. [Google Scholar] [CrossRef] [PubMed]
8. Kaur, T.; Diwakar, A.; Mirpuri, P.; Tripathi, M.; Chandra, P.S.; Gandhi, T.K. Artificial Intelligence in Epilepsy. *Neurol. India* **2021**, *69*, 560. [Google Scholar] [CrossRef]
9. Sohail, A. Genetic Algorithms in the Fields of Artificial Intelligence and Data Sciences. *Ann. Data Sci.* **2023**, *10*, 1007–1018. [Google Scholar] [CrossRef]
10. Lee, S.; Kim, H.-S. Prospect of Artificial Intelligence Based on Electronic Medical Record. *J. Lipid Atheroscler.* **2021**, *10*, 282–290. [Google Scholar] [CrossRef] [PubMed]
11. Contreras, I.; Vehi, J. Artificial Intelligence for Diabetes Management and Decision Support: Literature Review. *J. Med. Internet Res.* **2018**, *20*, e10775. [Google Scholar] [CrossRef]
12. Davis, C.R.; Murphy, K.J.; Curtis, R.G.; Maher, C.A. A Process Evaluation Examining the Performance, Adherence, and Acceptability of a Physical Activity and Diet Artificial Intelligence Virtual Health Assistant. *Int. J. Environ. Res. Public Health* **2020**, *17*, 9137. [Google Scholar] [CrossRef]
13. Bhatt, T.K.; Nimesh, S. *The Design and Development of Novel Drugs and Vaccines: Principles and Protocols*; Academic Press: Cambridge, MA, USA, 2021; ISBN 978-0-12-821475-6. [Google Scholar]
14. Paul, D.; Sanap, G.; Shenoy, S.; Kalyane, D.; Kalia, K.; Tekade, R.K. Artificial intelligence in drug discovery and development. *Drug Discov. Today* **2021**, *26*, 80–93. [Google Scholar] [CrossRef]
15. Mathai, N.; Chen, Y.; Kirchmair, J. Validation strategies for target prediction methods. *Brief. Bioinform.* **2020**, *21*, 791–802. [Google Scholar] [CrossRef]
16. Gupta, R.; Srivastava, D.; Sahu, M.; Tiwari, S.; Ambasta, R.K.; Kumar, P. Artificial intelligence to deep learning: Machine intelligence approach for drug discovery. *Mol. Divers.* **2021**, *25*, 1315–1360. [Google Scholar] [CrossRef]
17. Yaseen, B.T.; Kurnaz, S. Drug–target interaction prediction using artificial intelligence. *Appl. Nanosci.* **2023**, *13*, 3335–3345. [Google Scholar] [CrossRef]
18. Kumar, R.; Sharma, A.; Siddiqui, M.H.; Tiwari, R.K. Prediction of Drug-Plasma Protein Binding Using Artificial Intelligence Based Algorithms. *Comb. Chem. High Throughput Screen.* **2018**, *21*, 57–64. [Google Scholar] [CrossRef] [PubMed]

19. Romeo-Guitart, D.; Forés, J.; Herrando-Grabulosa, M.; Valls, R.; Leiva-Rodríguez, T.; Galea, E.; González-Pérez, F.; Navarro, X.; Petegnief, V.; Bosch, A.; et al. Neuroprotective Drug for Nerve Trauma Revealed Using Artificial Intelligence. *Sci. Rep.* **2018**, *8*, 1879. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
20. Rifaioğlu, A.S.; Atas, H.; Martin, M.J.; Cetin-Atalay, R.; Atalay, V.; Doğan, T. Recent applications of deep learning and machine intelligence on in silico drug discovery: Methods, tools and databases. *Brief. Bioinform.* **2019**, *20*, 1878–1912. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
21. Zhang, O.; Zhang, J.; Jin, J.; Zhang, X.; Hu, R.; Shen, C.; Cao, H.; Du, H.; Kang, Y.; Deng, Y.; et al. ResGen is a pocket-aware 3D molecular generation model based on parallel multiscale modelling. *Nat. Mach. Intell.* **2023**, *5*, 1020–1030. [[Google Scholar](#)] [[CrossRef](#)]
22. Gentile, F.; Yaacoub, J.C.; Gleave, J.; Fernandez, M.; Ton, A.-T.; Ban, F.; Stern, A.; Cherkasov, A. Artificial intelligence-enabled virtual screening of ultra-large chemical libraries with deep docking. *Nat. Protoc.* **2022**, *17*, 672–697. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
23. Carpenter, K.A.; Huang, X. Machine Learning-based Virtual Screening and Its Applications to Alzheimer's Drug Discovery: A Review. *Curr. Pharm. Des.* **2018**, *24*, 3347–3358. [[Google Scholar](#)] [[CrossRef](#)]
24. Jiménez-Luna, J.; Grisoni, F.; Schneider, G. Drug discovery with explainable artificial intelligence. *Nat. Mach. Intell.* **2020**, *2*, 573–584. [[Google Scholar](#)] [[CrossRef](#)]
25. Selvaraj, C.; Chandra, I.; Singh, S.K. Artificial intelligence and machine learning approaches for drug design: Challenges and opportunities for the pharmaceutical industries. *Mol. Divers.* **2022**, *26*, 1893–1913. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
26. Sadybekov, A.V.; Katritch, V. Computational approaches streamlining drug discovery. *Nature* **2023**, *616*, 673–685. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
27. Boobier, S.; Hose, D.R.J.; Blacker, A.J.; Nguyen, B.N. Machine learning with physicochemical relationships: Solubility prediction in organic solvents and water. *Nat. Commun.* **2020**, *11*, 5753. [[Google Scholar](#)] [[CrossRef](#)]
28. Fagerholm, U.; Hellberg, S.; Spjuth, O. Advances in Predictions of Oral Bioavailability of Candidate Drugs in Man with New Machine Learning Methodology. *Molecules* **2021**, *26*, 2572. [[Google Scholar](#)] [[CrossRef](#)]
29. Zhang, L.; Zhang, H.; Ai, H.; Hu, H.; Li, S.; Zhao, J.; Liu, H. Applications of Machine Learning Methods in Drug Toxicity Prediction. *Curr. Top. Med. Chem.* **2018**, *18*, 987–997. [[Google Scholar](#)] [[CrossRef](#)]
30. Ashburn, T.T.; Thor, K.B. Drug repositioning: Identifying and developing new uses for existing drugs. *Nat. Rev. Drug Discov.* **2004**, *3*, 673–683. [[Google Scholar](#)] [[CrossRef](#)]
31. Rani, P.; Dutta, K.; Kumar, V. Artificial intelligence techniques for prediction of drug synergy in malignant diseases: Past, present, and future. *Comput. Biol. Med.* **2022**, *144*, 105334. [[Google Scholar](#)] [[CrossRef](#)]
32. Gaweda, A.E.; Aronoff, G.R.; Brier, M.E. Use of Artificial Intelligence/Machine Learning for Individualization of Drug Dosing in Dialysis Patients. In *Technological Advances in Care of Patients with Kidney Diseases*; Saggi, S.J., Salifu, M.O., Eds.; Springer International Publishing: Cham, Switzerland, 2022; pp. 179–187. [[Google Scholar](#)] [[CrossRef](#)]
33. Vadapalli, S.; Abdelhalim, H.; Zeeshan, S.; Ahmed, Z. Artificial intelligence and machine learning approaches using gene expression and variant data for personalized medicine. *Brief. Bioinform.* **2022**, *23*, bbac191. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
34. Vo, T.H.; Nguyen, N.T.K.; Kha, Q.H.; Le, N.Q.K. On the road to explainable AI in drug-drug interactions prediction: A systematic review. *Comput. Struct. Biotechnol. J.* **2022**, *20*, 2112–2123. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
35. Dara, S.; Dhamercherla, S.; Jadav, S.S.; Babu, C.M.; Ahsan, M.J. Machine Learning in Drug Discovery: A Review. *Artif. Intell. Rev.* **2022**, *55*, 1947–1999. [[Google Scholar](#)] [[CrossRef](#)]

36. Zeng, X.; Xiang, H.; Yu, L.; Wang, J.; Li, K.; Nussinov, R.; Cheng, F. Accurate prediction of molecular properties and drug targets using a self-supervised image representation learning framework. *Nat. Mach. Intell.* **2022**, *4*, 1004–1016. [[Google Scholar](#)] [[CrossRef](#)]
37. Brito-Pacheco, C.; Brito-Loeza, C.; Martin-Gonzalez, A. A regularized logistic regression based model for supervised learning. *J. Algorithms Comput. Technol.* **2020**, *14*, 1748302620971535. [[Google Scholar](#)] [[CrossRef](#)]
38. Pisner, D.A.; Schnyer, D.M. Chapter 6—Support vector machine. In *Machine Learning*; Mechelli, A., Vieira, S., Eds.; Academic Press: Cambridge, MA, USA, 2020; pp. 101–121. [[Google Scholar](#)] [[CrossRef](#)]
39. Smaldone, A.M.; Kyro, G.W.; Batista, V.S. Quantum Convolutional Neural Networks for Multi-Channel Supervised Learning. *arXiv* **2023**, arXiv:2305.18961. [[Google Scholar](#)] [[CrossRef](#)]
40. Kaur, M.; Mohta, A. A Review of Deep Learning with Recurrent Neural Network. In Proceedings of the 2019 International Conference on Smart Systems and Inventive Technology (ICSSIT), Tirunelveli, India, 27–29 November 2019; IEEE: Tirunelveli, India, 2019; pp. 460–465. [[Google Scholar](#)] [[CrossRef](#)]
41. Lim, H.; Chon, K.-W.; Kim, M.-S. Active learning using Generative Adversarial Networks for improving generalization and avoiding distractor points. *Expert Syst. Appl.* **2023**, *227*, 120193. [[Google Scholar](#)] [[CrossRef](#)]
42. Sinaga, K.P.; Yang, M.-S. Unsupervised K-Means Clustering Algorithm. *IEEE Access* **2020**, *8*, 80716–80727. [[Google Scholar](#)] [[CrossRef](#)]
43. Lakshmi, B.S.S.S.; Ravi Kiran Varma, P. Machine Learning for Drug Discovery Using Agglomerative Hierarchical Clustering. In *Soft Computing and Signal Processing*; Reddy, V.S., Prasad, V.K., Wang, J., Reddy, K.T.V., Eds.; Smart Innovation, Systems and Technologies Book Series; Springer Nature: Singapore, 2023; pp. 127–137. [[Google Scholar](#)] [[CrossRef](#)]
44. Lever, J.; Krzywinski, M.; Altman, N. Points of Significance: Principal component analysis. *Nat. Methods* **2017**, *14*, 641–643. [[Google Scholar](#)] [[CrossRef](#)]
45. Balamurali, M. t-Distributed Stochastic Neighbor Embedding. In *Encyclopedia of Mathematical Geosciences*; Daya Sagar, B.S., Cheng, Q., McKinley, J., Agterberg, F., Eds.; Encyclopedia of Earth Sciences Series; Springer International Publishing: Cham, Switzerland, 2020; pp. 1–9. [[Google Scholar](#)] [[CrossRef](#)]
46. Rickert, C.A.; Lieleg, O. Machine learning approaches for biomolecular, biophysical, and biomaterials research. *Biophys. Rev.* **2022**, *3*, 021306. [[Google Scholar](#)] [[CrossRef](#)]
47. Stern, M.; Arinze, C.; Perez, L.; Palmer, S.E.; Murugan, A. Supervised learning through physical changes in a mechanical system. *Proc. Natl. Acad. Sci. USA* **2020**, *117*, 14843–14850. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
48. Usama, M.; Qadir, J.; Raza, A.; Arif, H.; Yau, K.A.; Elkhatab, Y.; Hussain, A.; Al-Fuqaha, A. Unsupervised Machine Learning for Networking: Techniques, Applications and Research Challenges. *IEEE Access* **2019**, *7*, 65579–65615. [[Google Scholar](#)] [[CrossRef](#)]
49. Talevi, A.; Morales, J.F.; Hather, G.; Podichetty, J.T.; Kim, S.; Bloomingdale, P.C.; Kim, S.; Burton, J.; Brown, J.D.; Winterstein, A.G.; et al. Machine Learning in Drug Discovery and Development Part 1: A Primer. *CPT Pharmacomet. Syst. Pharmacol.* **2020**, *9*, 129–142. [[Google Scholar](#)] [[CrossRef](#)]
50. Zhou, Z.; Li, X.; Zare, R.N. Optimizing Chemical Reactions with Deep Reinforcement Learning. *ACS Cent. Sci.* **2017**, *3*, 1337–1344. [[Google Scholar](#)] [[CrossRef](#)]
51. Kramer, O. Scikit-Learn. In *Machine Learning for Evolution Strategies*; Kramer, O., Ed.; Studies in Big Data; Springer International Publishing: Cham, Switzerland, 2016; pp. 45–53. [[Google Scholar](#)] [[CrossRef](#)]
52. Imambi, S.; Prakash, K.B.; Kanagachidambaresan, G.R. PyTorch. In *Programming with TensorFlow: Solution for Edge Computing Applications*; Prakash, K.B.,

- Kanagachidambaresan, G.R., Eds.; EAI/Springer Innovations in Communication and Computing; Springer International Publishing: Cham, Switzerland, 2021; pp. 87–104. [[Google Scholar](#)] [[CrossRef](#)]
53. Gulli, A.; Pal, S. *Deep Learning with Keras*; Packt Publishing Ltd.: Birmingham, UK, 2017. [[Google Scholar](#)]
54. Priya, S.; Tripathi, G.; Singh, D.B.; Jain, P.; Kumar, A. Machine learning approaches and their applications in drug discovery and design. *Chem. Biol. Drug Des.* **2022**, *100*, 136–153. [[Google Scholar](#)] [[CrossRef](#)]
55. Pandis, N. Linear regression. *Am. J. Orthod. Dentofac. Orthop.* **2016**, *149*, 431–434. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
56. Taskinen, J.; Yliruusi, J. Prediction of physicochemical properties based on neural network modelling. *Adv. Drug Deliv. Rev.* **2003**, *55*, 1163–1183. [[Google Scholar](#)] [[CrossRef](#)]
57. Suprpto, S.; Nikmah, Y.L. Ridge and Lasso Regression for Feature Selection of Overlapping Ibuprofen and Paracetamol UV Spectra. *Moroc. J. Chem.* **2023**, *11*, 11–229. [[Google Scholar](#)] [[CrossRef](#)]
58. Roozbeh, M.; Arashi, M.; Hamzah, N.A. Generalized Cross-Validation for Simultaneous Optimization of Tuning Parameters in Ridge Regression. *Iran. J. Sci. Technol. Trans. Sci.* **2020**, *44*, 473–485. [[Google Scholar](#)] [[CrossRef](#)]
59. Kim, Y.; Hao, J.; Mallavarapu, T.; Park, J.; Kang, M. Hi-LASSO: High-Dimensional LASSO. *IEEE Access* **2019**, *7*, 44562–44573. [[Google Scholar](#)] [[CrossRef](#)]
60. Abdulhafedh, A. Comparison between Common Statistical Modeling Techniques Used in Research, Including: Discriminant Analysis vs Logistic Regression, Ridge Regression vs LASSO, and Decision Tree vs Random Forest. *Open Access Libr. J.* **2022**, *9*, 1–19. [[Google Scholar](#)] [[CrossRef](#)]
61. Patel, L.; Shukla, T.; Huang, X.; Ussery, D.W.; Wang, S. Machine Learning Methods in Drug Discovery. *Molecules* **2020**, *25*, 5277. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
62. Peña-Guerrero, J.; Nguewa, P.A.; García-Sosa, A.T. Machine learning, artificial intelligence, and data science breaking into drug design and neglected diseases. *WIREs Comput. Mol. Sci.* **2021**, *11*, e1513. [[Google Scholar](#)] [[CrossRef](#)]
63. Ozdemir, S.; Susarla, D. *Feature Engineering Made Easy: Identify Unique Features from Your Dataset in Order to Build Powerful Machine Learning Systems*; Packt Publishing Ltd.: Birmingham, UK, 2018. [[Google Scholar](#)]
64. Chauhan, N.K.; Singh, K. A Review on Conventional Machine Learning vs Deep Learning. In Proceedings of the 2018 International Conference on Computing, Power and Communication Technologies (GUCON), Greater Noida, India, 28–29 September 2018; IEEE: Greater Noida, India, 2018; pp. 347–352. [[Google Scholar](#)] [[CrossRef](#)]
65. Janiesch, C.; Zschech, P.; Heinrich, K. Machine learning and deep learning. *Electron. Mark.* **2021**, *31*, 685–695. [[Google Scholar](#)] [[CrossRef](#)]
66. Gong, D.; Ben-Akiva, E.; Singh, A.; Yamagata, H.; Est-Witte, S.; Shade, J.K.; Trayanova, N.A.; Green, J.J. Machine learning guided structure function predictions enable in silico nanoparticle screening for polymeric gene delivery. *Acta Biomater.* **2022**, *154*, 349–358. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
67. Taye, M.M. Understanding of Machine Learning with Deep Learning: Architectures, Workflow, Applications and Future Directions. *Computers* **2023**, *12*, 91. [[Google Scholar](#)] [[CrossRef](#)]
68. Choudhary, K.; Kalish, I.; Beams, R.; Tavazza, F. High-throughput Identification and Characterization of Two-dimensional Materials using Density functional theory. *Sci. Rep.* **2017**, *7*, 5179. [[Google Scholar](#)] [[CrossRef](#)]
69. Friedman, R. Computational studies of protein–drug binding affinity changes upon mutations in the drug target. *WIREs Comput. Mol. Sci.* **2022**, *12*, e1563. [[Google Scholar](#)] [[CrossRef](#)]

70. Varadharajan, V.; Arumugam, G.S.; Shanmugam, S. Isatin-based virtual high throughput screening, molecular docking, DFT, QM/MM, MD and MM-PBSA study of novel inhibitors of SARS-CoV-2 main protease. *J. Biomol. Struct. Dyn.* **2022**, *40*, 7852–7867. [[Google Scholar](#)] [[CrossRef](#)]
71. Joseph Sahayarayan, J.; Soundar Rajan, K.; Nachiappan, M.; Prabhu, D.; Guru Raj Rao, R.; Jeyakanthan, J.; Hossam Mahmoud, A.; Mohammed, O.B.; Morgan, A.M.A. Identification of potential drug target in malarial disease using molecular docking analysis. *Saudi J. Biol. Sci.* **2020**, *27*, 3327–3333. [[Google Scholar](#)] [[CrossRef](#)]
72. Faris, A.; Ibrahim, I.M.; Hadni, H.; Elhallaoui, M. High-throughput virtual screening of phenylpyrimidine derivatives as selective JAK3 antagonists using computational methods. *J. Biomol. Struct. Dyn.* **2023**, 1–26. [[Google Scholar](#)] [[CrossRef](#)]
73. Sohlenius-Sternbeck, A.-K.; Terelius, Y. Evaluation of ADMET Predictor in Early Discovery Drug Metabolism and Pharmacokinetics Project Work. *Drug Metab. Dispos.* **2022**, *50*, 95–104. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
74. Öeren, M.; Walton, P.J.; Hunt, P.A.; Ponting, D.J.; Segall, M.D. Predicting reactivity to drug metabolism: Beyond P450s—Modelling FMOs and UGTs. *J. Comput. Aided Mol. Des.* **2021**, *35*, 541–555. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
75. Rydberg, P.; Jørgensen, F.S.; Olsen, L. Use of density functional theory in drug metabolism studies. *Expert Opin. Drug Metab. Toxicol.* **2014**, *10*, 215–227. [[Google Scholar](#)] [[CrossRef](#)]
76. Smirnova, A.; Yablonskiy, M.; Petrov, V.; Mitrofanov, A. DFT Prediction of Radiolytic Stability of Conformationally Flexible Ligands. *Energies* **2023**, *16*, 257. [[Google Scholar](#)] [[CrossRef](#)]
77. Alzain, A.A.; Elbadwi, F.A.; Alsamani, F.O. Discovery of novel TMPRSS2 inhibitors for COVID-19 using in silico fragment-based drug design, molecular docking, molecular dynamics, and quantum mechanics studies. *Inform. Med. Unlocked* **2022**, *29*, 100870. [[Google Scholar](#)] [[CrossRef](#)]
78. Bhatnagar, R.; Sardar, S.; Beheshti, M.; Podichetty, J.T. How can natural language processing help model informed drug development?: A review. *JAMIA Open* **2022**, *5*, ooac043. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
79. Zheng, S.; Dharssi, S.; Wu, M.; Li, J.; Lu, Z. Text Mining for Drug Discovery. In *Bioinformatics and Drug Discovery*; Larson, R.S., Oprea, T.I., Eds.; Methods in Molecular Biology; Springer: New York, NY, USA, 2019; pp. 231–252. [[Google Scholar](#)] [[CrossRef](#)]
80. Dumitriu, A.; Molony, C.; Daluwatte, C. Graph-Based Natural Language Processing for the Pharmaceutical Industry. In *Provenance in Data Science: From Data Models to Context-Aware Knowledge Graphs*; Sikos, L.F., Seneviratne, O.W., McGuinness, D.L., Eds.; Advanced Information and Knowledge Processing; Springer International Publishing: Cham, Switzerland, 2021; pp. 75–110. [[Google Scholar](#)] [[CrossRef](#)]
81. Study of the Drug-Related Adverse Events with the Help of Electronic Health Records and Natural Language Processing—ProQuest. Available online: <https://www.proquest.com/openview/e0e053ffe5b850bd656912f47db18b77/1?pq-origsite=gscholar&cbl=5444811> (accessed on 23 October 2023).
82. Corcoran, C.M.; Mittal, V.A.; Bearden, C.E.; Gur, R.E.; Hitczenko, K.; Bilgrami, Z.; Savic, A.; Cecchi, G.A.; Wolff, P. Language as a biomarker for psychosis: A natural language processing approach. *Schizophr. Res.* **2020**, *226*, 158–166. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
83. Jang, G.; Lee, T.; Hwang, S.; Park, C.; Ahn, J.; Seo, S.; Hwang, Y.; Yoon, Y. PISTON: Predicting drug indications and side effects using topic modeling and natural language processing. *J. Biomed. Inform.* **2018**, *87*, 96–107. [[Google Scholar](#)] [[CrossRef](#)]
84. Piñero, J.; Bravo, À.; Queralt-Rosinach, N.; Gutiérrez-Sacristán, A.; Deu-Pons, J.; Centeno, E.; García-García, J.; Sanz, F.; Furlong, L.I. DisGeNET: A comprehensive platform

- integrating information on human disease-associated genes and variants. *Nucleic Acids Res.* **2017**, *45*, D833–D839. [Google Scholar] [CrossRef]
85. Szklarczyk, D.; Gable, A.L.; Lyon, D.; Junge, A.; Wyder, S.; Huerta-Cepas, J.; Simonovic, M.; Doncheva, N.T.; Morris, J.H.; Bork, P.; et al. STRING v11: Protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Res.* **2019**, *47*, D607–D613. [Google Scholar] [CrossRef]
86. Szklarczyk, D.; Santos, A.; von Mering, C.; Jensen, L.J.; Bork, P.; Kuhn, M. STITCH 5: Augmenting protein–chemical interaction networks with tissue and affinity data. *Nucleic Acids Res.* **2016**, *44*, D380–D384. [Google Scholar] [CrossRef]
87. Salehi, P.; Chalechale, A.; Taghizadeh, M. Generative Adversarial Networks (GANs): An Overview of Theoretical Model, Evaluation Metrics, and Recent Developments. *arXiv* **2020**, arXiv:2005.13178. [Google Scholar] [CrossRef]
88. Blanchard, A.E.; Stanley, C.; Bhowmik, D. Using GANs with adaptive training data to search for new molecules. *J. Cheminformatics* **2021**, *13*, 14. [Google Scholar] [CrossRef]
89. Abbasi, M.; Santos, B.P.; Pereira, T.C.; Sofia, R.; Monteiro, N.R.C.; Simões, C.J.V.; Brito, R.M.M.; Ribeiro, B.; Oliveira, J.L.; Arrais, J.P. Designing optimized drug candidates with Generative Adversarial Network. *J. Cheminformatics* **2022**, *14*, 40. [Google Scholar] [CrossRef]
90. Zhang, Z.; Li, F.; Guan, J.; Kong, Z.; Shi, L.; Zhou, S. GANs for Molecule Generation in Drug Design and Discovery. In *Generative Adversarial Learning: Architectures and Applications*; Razavi-Far, R., Ruiz-Garcia, A., Palade, V., Schmidhuber, J., Eds.; Intelligent Systems Reference Library; Springer International Publishing: Cham, Switzerland, 2022; pp. 233–273. [Google Scholar] [CrossRef]
91. Barcelos, M.P.; Gomes, S.Q.; Federico, L.B.; Francischini, I.A.G.; Hage-Melim, L.I.d.S.; Silva, G.M.; de Paula da Silva, C.H.T. Lead Optimization in Drug Discovery. In *Research Topics in Bioactivity, Environment and Energy: Experimental and Theoretical Tools*; Taft, C.A., de Lazaro, S.R., Eds.; Engineering Materials; Springer International Publishing: Cham, Switzerland, 2022; pp. 481–500. [Google Scholar] [CrossRef]
92. Lin, E.; Lin, C.-H.; Lane, H.-Y. Relevant Applications of Generative Adversarial Networks in Drug Design and Discovery: Molecular De Novo Design, Dimensionality Reduction, and De Novo Peptide and Protein Design. *Molecules* **2020**, *25*, 3250. [Google Scholar] [CrossRef]
93. Wu, B.; Li, L.; Cui, Y.; Zheng, K. Cross-Adversarial Learning for Molecular Generation in Drug Design. *Front. Pharmacol.* **2022**, *12*, 827606. Available online: <https://www.frontiersin.org/articles/10.3389/fphar.2021.827606> (accessed on 20 October 2023). [CrossRef] [PubMed]
94. Zhao, L.; Wang, J.; Pang, L.; Liu, Y.; Zhang, J. GANsDTA: Predicting Drug-Target Binding Affinity Using GANs. *Front. Genet.* **2020**, *10*, 1243. Available online: <https://www.frontiersin.org/articles/10.3389/fgene.2019.01243> (accessed on 20 October 2023). [CrossRef] [PubMed]
95. Gan, J.; Liu, J.; Liu, Y.; Chen, S.; Dai, W.; Xiao, Z.-X.; Cao, Y. DrugRep: An automatic virtual screening server for drug repurposing. *Acta Pharmacol. Sin.* **2023**, *44*, 888–896. [Google Scholar] [CrossRef] [PubMed]
96. Tripathi, S.; Augustin, A.I.; Dunlop, A.; Sukumaran, R.; Dheer, S.; Zavalny, A.; Haslam, O.; Austin, T.; Donchez, J.; Tripathi, P.K.; et al. Recent advances and application of generative adversarial networks in drug discovery, development, and targeting. *Artif. Intell. Life Sci.* **2022**, *2*, 100045. [Google Scholar] [CrossRef]
97. Sureyya Rifaioglu, A.; Nalbat, E.; Atalay, V.; Jesus Martin, M.; Cetin-Atalay, R.; Doğan, T. DEEPScreen: High performance drug–target interaction prediction with convolutional neural networks using 2-D structural compound representations. *Chem. Sci.* **2020**, *11*, 2531–2557. [Google Scholar] [CrossRef]

98. Cai, C.; Wang, S.; Xu, Y.; Zhang, W.; Tang, K.; Ouyang, Q.; Lai, L.; Pei, J. Transfer Learning for Drug Discovery. *J. Med. Chem.* **2020**, *63*, 8683–8694. [[Google Scholar](#)] [[CrossRef](#)]
99. Zhuang, D.; Ibrahim, A.K. Deep Learning for Drug Discovery: A Study of Identifying High Efficacy Drug Compounds Using a Cascade Transfer Learning Approach. *Appl. Sci.* **2021**, *11*, 7772. [[Google Scholar](#)] [[CrossRef](#)]
100. Miller, E.; Hansch, C. Structure-Activity Analysis of Tetrahydrofolate Analogs Using Substituent Constants and Regression Analysis. *J. Pharm. Sci.* **1967**, *56*, 92–97. [[Google Scholar](#)] [[CrossRef](#)]
101. Schneider, N.; Jäckels, C.; Andres, C.; Hutter, M.C. Gradual in Silico Filtering for Druglike Substances. *J. Chem. Inf. Model.* **2008**, *48*, 613–628. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
102. Jorissen, R.N.; Gilson, M.K. Virtual Screening of Molecular Databases Using a Support Vector Machine. *J. Chem. Inf. Model.* **2005**, *45*, 549–561. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
103. Hou, T.; Wang, J.; Li, Y. ADME Evaluation in Drug Discovery. 8. The Prediction of Human Intestinal Absorption by a Support Vector Machine. *J. Chem. Inf. Model.* **2007**, *47*, 2408–2415. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
104. Wu, Z.; Ramsundar, B.; Feinberg, E.N.; Gomes, J.; Geniesse, C.; Pappu, A.S.; Leswing, K.; Pande, V. MoleculeNet: A benchmark for molecular machine learning. *Chem. Sci.* **2018**, *9*, 513–530. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
105. Ma, J.; Sheridan, R.P.; Liaw, A.; Dahl, G.E.; Svetnik, V. Deep Neural Nets as a Method for Quantitative Structure–Activity Relationships. *J. Chem. Inf. Model.* **2015**, *55*, 263–274. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
106. Yang, K.; Swanson, K.; Jin, W.; Coley, C.; Eiden, P.; Gao, H.; Guzman-Perez, A.; Hopper, T.; Kelley, B.; Mathea, M.; et al. Analyzing Learned Molecular Representations for Property Prediction. *J. Chem. Inf. Model.* **2019**, *59*, 3370–3388. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
107. Ragoza, M.; Hochuli, J.; Idrobo, E.; Sunseri, J.; Koes, D.R. Protein–Ligand Scoring with Convolutional Neural Networks. *J. Chem. Inf. Model.* **2017**, *57*, 942–957. [[Google Scholar](#)] [[CrossRef](#)]
108. Wang, C.; Zhang, Y. Improving scoring-docking-screening powers of protein–ligand scoring functions using random forest. *J. Comput. Chem.* **2017**, *38*, 169–177. [[Google Scholar](#)] [[CrossRef](#)]
109. Segler, M.H.S.; Preuss, M.; Waller, M.P. Learning to Plan Chemical Syntheses. *Nature* **2018**, *555*, 604–610. [[Google Scholar](#)] [[CrossRef](#)]
110. Coley, C.W.; Thomas, D.A.; Lummiss, J.A.M.; Jaworski, J.N.; Breen, C.P.; Schultz, V.; Hart, T.; Fishman, J.S.; Rogers, L.; Gao, H.; et al. A robotic platform for flow synthesis of organic compounds informed by AI planning. *Science* **2019**, *365*, eaax1566. [[Google Scholar](#)] [[CrossRef](#)]
111. Xu, Y.; Lin, K.; Wang, S.; Wang, L.; Cai, C.; Song, C.; Lai, L.; Pei, J. Deep learning for molecular generation. *Future Med. Chem.* **2019**, *11*, 567–597. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
112. Elton, D.C.; Boukouvalas, Z.; Fuge, M.D.; Chung, P.W. Deep learning for molecular design—A review of the state of the art. *Mol. Syst. Des. Eng.* **2019**, *4*, 828–849. [[Google Scholar](#)] [[CrossRef](#)]
113. Dana, D.; Gadhiya, S.V.; St. Surin, L.G.; Li, D.; Naaz, F.; Ali, Q.; Paka, L.; Yamin, M.A.; Narayan, M.; Goldberg, I.D.; et al. Deep Learning in Drug Discovery and Medicine; Scratching the Surface. *Molecules* **2018**, *23*, 2384. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
114. Kitchin, R.; McArdle, G. What makes Big Data, Big Data? Exploring the ontological characteristics of 26 datasets. *Big Data Soc.* **2016**, *3*, 2053951716631130. [[Google Scholar](#)] [[CrossRef](#)]

115. Reker, D.; Schneider, G. Active-learning strategies in computer-assisted drug discovery. *Drug Discov. Today* **2015**, *20*, 458–465. [Google Scholar] [CrossRef] [PubMed]
116. DeepChem. Available online: <https://github.com/deepchem/deepchem> (accessed on 20 October 2023).
117. Merck/DeepNeuralNet-QSAR. Merck Sharp & Dohme Corp. a Subsidiary of Merck & Co., Inc. Available online: <https://github.com/Merck/DeepNeuralNet-QSAR> (accessed on 20 October 2023).
118. Keenan, G. *Croningp/ChemputerSoftware: Chemputer First Release*; Version 0.1.1; Zenodo: Genève, Switzerland, 2018. [Google Scholar] [CrossRef]
119. Mayr, A.; Klambauer, G.; Unterthiner, T.; Hochreiter, S. DeepTox: Toxicity Prediction using Deep Learning. *Front. Environ. Sci.* **2016**, *3*, 80. [Google Scholar] [CrossRef]
120. AlphaFold. Available online: <https://www.deepmind.com/research/highlighted-research/alphafold> (accessed on 20 October 2023).
121. Jumper, J.; Evans, R.; Pritzel, A.; Green, T.; Figurnov, M.; Ronneberger, O.; Tunyasuvunakool, K.; Bates, R.; Židek, A.; Potapenko, A.; et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **2021**, *596*, 583–589. [Google Scholar] [CrossRef] [PubMed]
122. ORGANIC. The Matter Lab, Aspuru-Guzik Group Repo, 80 St. George Street Toronto, ON, M5S 3H6. Available online: <https://github.com/aspuru-guzik-group/ORGANIC> (accessed on 20 October 2023).
123. Kwak, H.S.; An, Y.; Giesen, D.J.; Hughes, T.F.; Brown, C.T.; Leswing, K.; Abroshan, H.; Halls, M.D. Design of Organic Electronic Materials with a Goal-Directed Generative Model Powered by Deep Neural Networks and High-Throughput Molecular Simulations. *Front. Chem.* **2022**, *9*, 800370. Available online: <https://www.frontiersin.org/articles/10.3389/fchem.2021.800370> (accessed on 20 October 2023). [CrossRef] [PubMed]
124. Mostaghimi, M.; Rêgo, C.R.C.; Haldar, R.; Wöll, C.; Wenzel, W.; Kozłowska, M. Automated Virtual Design of Organic Semiconductors Based on Metal–Organic Frameworks. *Front. Mater.* **2022**, *9*, 840644. Available online: <https://www.frontiersin.org/articles/10.3389/fmats.2022.840644> (accessed on 20 October 2023). [CrossRef]
125. Chen, P.; Wang, Y.; Yan, H.; Gao, S.; Xu, Z.; Li, Y.; Mo, Q.; Huang, J.; Tao, J.; Pan, G.; et al. 3DStructGen: An interactive web-based 3D structure generation for non-periodic molecule and crystal. *J. Cheminform.* **2020**, *12*, 7. [Google Scholar] [CrossRef] [PubMed]
126. Feinberg, E.N.; Sur, D.; Wu, Z.; Husic, B.E.; Mai, H.; Li, Y.; Sun, S.; Yang, J.; Ramsundar, B.; Pande, V.S. PotentialNet for Molecular Property Prediction. *ACS Cent. Sci.* **2018**, *4*, 1520–1530. [Google Scholar] [CrossRef] [PubMed]
127. Seo, S.; Choi, J.; Park, S.; Ahn, J. Binding affinity prediction for protein–ligand complex using deep attention mechanism based on intermolecular interactions. *BMC Bioinform.* **2021**, *22*, 542. [Google Scholar] [CrossRef] [PubMed]
128. Kandel, J.; Tayara, H.; Chong, K.T. PURESNet: Prediction of protein–ligand binding sites using deep residual neural network. *J. Cheminform.* **2021**, *13*, 65. [Google Scholar] [CrossRef]
129. Wang, C. DeltaVina. Available online: <https://github.com/chengwang88/deltavina> (accessed on 20 October 2023).
130. Kumar, S.; Kim, M. SMPLIP-Score: Predicting ligand binding affinity from simple and interpretable on-the-fly interaction fingerprint pattern descriptors. *J. Cheminform.* **2021**, *13*, 28. [Google Scholar] [CrossRef]
131. Yang, C.; Zhang, Y. Delta Machine Learning to Improve Scoring–Ranking–Screening Performances of Protein–Ligand Scoring Functions. *J. Chem. Inf. Model.* **2022**, *62*, 2696–2712. [Google Scholar] [CrossRef]

132. Neural Graph Fingerprints. Formerly: Harvard Intelligent Probabilistic Systems Group—Now at Princeton. Available online: <https://github.com/HIPS/neural-fingerprint> (accessed on 20 October 2023).
133. Wen, N.; Liu, G.; Zhang, J.; Zhang, R.; Fu, Y.; Han, X. A fingerprints based molecular property prediction method using the BERT model. *J. Cheminform.* **2022**, *14*, 71. [[Google Scholar](#)] [[CrossRef](#)]
134. GastroPlus® PBPK & PBBM Modeling and Simulation. Simulations Plus. Available online: <https://www.simulations-plus.com/software/gastroplus/> (accessed on 20 October 2023).
135. Hussain, A.; Afzal, O.; Yasmin, S.; Haider, N.; Altamimi, A.S.A.; Martinez, F.; Acree, W.E., Jr.; Ramzan, M. Preferential Solvation Study of Rosuvastatin in the {PEG400 (1) + Water (2)} Cosolvent Mixture and GastroPlus Software-Based In Vivo Predictions. *ACS Omega* **2023**, *8*, 12761–12772. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
136. Schneider, P.; Walters, W.P.; Plowright, A.T.; Sieroka, N.; Listgarten, J.; Goodnow, R.A.; Fisher, J.; Jansen, J.M.; Duca, J.S.; Rush, T.S.; et al. Rethinking drug design in the artificial intelligence era. *Nat. Rev. Drug Discov.* **2020**, *19*, 353–364. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
137. Ertl, P.; Schuffenhauer, A. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *J. Cheminform.* **2009**, *1*, 8. [[Google Scholar](#)] [[CrossRef](#)]
138. Chaudhari, R.; Fong, L.W.; Tan, Z.; Huang, B.; Zhang, S. An up-to-date overview of computational polypharmacology in modern drug discovery. *Expert Opin. Drug Discov.* **2020**, *15*, 1025–1044. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
139. Reddy, A.S.; Zhang, S. Polypharmacology: Drug discovery for the future. *Expert Rev. Clin. Pharmacol.* **2013**, *6*, 41–47. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
140. Sirois, C.; Khoury, R.; Durand, A.; Deziel, P.-L.; Bukhtiyarova, O.; Chiu, Y.; Talbot, D.; Bureau, A.; Després, P.; Gagné, C.; et al. Exploring polypharmacy with artificial intelligence: Data analysis protocol. *BMC Med. Inform. Decis. Mak.* **2021**, *21*, 219. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
141. Duch, W.; Swaminathan, K.; Meller, J. Artificial Intelligence Approaches for Rational Drug Design and Discovery. *Curr. Pharm. Des.* **2007**, *13*, 1497–1508. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
142. Blasiak, A.; Khong, J.; Kee, T. CURATE.AI: Optimizing Personalized Medicine with Artificial Intelligence. *SLAS Technol. Transl. Life Sci. Innov.* **2020**, *25*, 95–105. [[Google Scholar](#)] [[CrossRef](#)]
143. Baronzio, G.; Parmar, G.; Baronzio, M. Overview of Methods for Overcoming Hindrance to Drug Delivery to Tumors, with Special Attention to Tumor Interstitial Fluid. *Front. Oncol.* **2015**, *5*, 165. Available online: <https://www.frontiersin.org/articles/10.3389/fonc.2015.00165> (accessed on 20 October 2023). [[CrossRef](#)]
144. Nexocode. Available online: <https://nexocode.com/> (accessed on 20 October 2023).
145. Universität Basel. Available online: <https://www.unibas.ch/de.html> (accessed on 20 October 2023).
146. Popular Mechanics—Product Reviews, How-To, Space, Military, Math, Science, and New Technology. Popular Mechanics. Available online: <https://www.popularmechanics.com/> (accessed on 20 October 2023).
147. IBM—United Kingdom. Available online: <https://www.ibm.com/uk-en> (accessed on 20 October 2023).
148. Main Page. Wikipedia, the Free Encyclopedia. Available online: https://en.wikipedia.org/w/index.php?title=Main_Page&oldid=1114291180 (accessed on 20 October 2023).

149. Zong, N.; Wen, A.; Moon, S.; Fu, S.; Wang, L.; Zhao, Y.; Yu, Y.; Huang, M.; Wang, Y.; Zheng, G.; et al. Computational drug repurposing based on electronic health records: A scoping review. *Npj Digit. Med.* **2022**, *5*, 77. [[Google Scholar](#)] [[CrossRef](#)]
150. Zhou, Y.; Wang, F.; Tang, J.; Nussinov, R.; Cheng, F. Artificial intelligence in COVID-19 drug repurposing. *Lancet Digit. Health* **2020**, *2*, e667–e676. [[Google Scholar](#)] [[CrossRef](#)]
151. Wang, Y.; Aldahdooh, J.; Hu, Y.; Yang, H.; Vähä-Koskela, M.; Tang, J.; Tanoli, Z. DrugRepo: A novel approach to repurposing drugs based on chemical and genomic features. *Sci. Rep.* **2022**, *12*, 21116. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
152. Lejal, N.; Tarus, B.; Bouguyon, E.; Chenavas, S.; Bertho, N.; Delmas, B.; Ruigrok, R.W.H.; Di Primo, C.; Slama-Schwok, A. Structure-Based Discovery of the Novel Antiviral Properties of Naproxen against the Nucleoprotein of Influenza A Virus. *Antimicrob. Agents Chemother.* **2013**, *57*, 2231–2242. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
153. Terrier, O.; Dilly, S.; Pizzorno, A.; Chalupska, D.; Humpolickova, J.; Bouřa, E.; Berenbaum, F.; Quideau, S.; Lina, B.; Fève, B.; et al. Antiviral Properties of the NSAID Drug Naproxen Targeting the Nucleoprotein of SARS-CoV-2 Coronavirus. *Molecules* **2021**, *26*, 2593. [[Google Scholar](#)] [[CrossRef](#)]
154. Mostafa, A.; Kandeil, A.; Elshaier, Y.A.M.M.; Kutkat, O.; Moatasim, Y.; Rashad, A.A.; Shehata, M.; Gomaa, M.R.; Mahrous, N.; Mahmoud, S.H.; et al. FDA-Approved Drugs with Potent In Vitro Antiviral Activity against Severe Acute Respiratory Syndrome Coronavirus 2. *Pharmaceuticals* **2020**, *13*, 443. [[Google Scholar](#)] [[CrossRef](#)]
155. Stebbing, J.; Krishnan, V.; de Bono, S.; Ottaviani, S.; Casalini, G.; Richardson, P.J.; Monteil, V.; Lauschke, V.M.; Mirazimi, A.; Youhanna, S.; et al. Mechanism of baricitinib supports artificial intelligence-predicted testing in COVID-19 patients. *EMBO Mol. Med.* **2020**, *12*, e12697. [[Google Scholar](#)] [[CrossRef](#)]
156. Farghali, H.; Kutinová Canová, N.; Arora, M. The Potential Applications of Artificial Intelligence in Drug Discovery and Development. *Physiol. Res.* **2021**, *70* (Suppl. 4), S715–S722. [[Google Scholar](#)] [[CrossRef](#)]
157. Evaluation of Safety, Tolerability & PK of DSP-2230 in Healthy Subjects. Health Research Authority. Available online: <https://www.hra.nhs.uk/planning-and-improving-research/application-summaries/research-summaries/evaluation-of-safetytolerability-pk-of-dsp-2230-in-healthy-subjects/> (accessed on 20 October 2023).
158. Terranova, N.; Jansen, M.; Falk, M.; Hendriks, B.S. Population pharmacokinetics of ATR inhibitor berzosertib in phase I studies for different cancer types. *Cancer Chemother. Pharmacol.* **2021**, *87*, 185–196. [[Google Scholar](#)] [[CrossRef](#)]
159. Plummer, R.; Dean, E.; Arkenau, H.-T.; Redfern, C.; Spira, A.I.; Melear, J.M.; Chung, K.Y.; Ferrer-Playan, J.; Goddemeier, T.; Locatelli, G.; et al. A phase 1b study evaluating the safety and preliminary efficacy of berzosertib in combination with gemcitabine in patients with advanced non-small cell lung cancer. *Lung Cancer* **2022**, *163*, 19–26. [[Google Scholar](#)] [[CrossRef](#)]
160. Almallah, Z.; El-Lababidi, R.; Shamout, F.; Doyle, D.J. Artificial Intelligence: The New Alexander Fleming. *Healthc. Inform. Res.* **2021**, *27*, 168–171. [[Google Scholar](#)] [[CrossRef](#)] [[PubMed](#)]
161. Valavanidis, A. Artificial Intelligence Application with Machine-Learning Algorithm Identified a Powerful Broad-Spectrum Antibiotic. Available online: <http://chem-tox-ecotox.org/wp-content/uploads/2020/03/ANTIBIOTICS-HALICIN-ARTIFICIAL-INTELLIGENCE-2020.pdf> (accessed on 2 January 2024).
162. Markham, A. Evinacumab: First Approval. *Drugs* **2021**, *81*, 1101–1105. [[Google Scholar](#)] [[CrossRef](#)]
163. Miller, M.; Tokgozoglu, L.; Parhofer, K.G.; Handelsman, Y.; Leiter, L.A.; Landmesser, U.; Brinton, E.A.; Catapano, A.L. Icosapent ethyl for reduction of persistent cardiovascular risk: A

- critical review of major medical society guidelines and statements. *Expert Rev. Cardiovasc. Ther.* **2022**, *20*, 609–625. [Google Scholar] [CrossRef]
164. Ballantyne, C.M.; Manku, M.S.; Bays, H.E.; Philip, S.; Granowitz, C.; Doyle, R.T.; Juliano, R.A. Icosapent Ethyl Effects on Fatty Acid Profiles in Statin-Treated Patients with High Triglycerides: The Randomized, Placebo-controlled ANCHOR Study. *Cardiol. Ther.* **2019**, *8*, 79–90. [Google Scholar] [CrossRef] [PubMed]
165. kgi-admin. Delpazolid by LegoChem Biosciences for Tuberculosis: Likelihood of Approval. Pharmaceutical Technology. Available online: <https://www.pharmaceutical-technology.com/data-insights/delpazolid-legochem-biosciences-tuberculosis-likelihood-of-approval/> (accessed on 20 October 2023).
166. Healthcare, G. First Drug Created by AI Enters Clinical Trials. Clinical Trials Arena. Available online: <https://www.clinicaltrialsarena.com/comment/first-drug-created-ai-enters-trials/> (accessed on 20 October 2023).
167. Evaxion Cleared by FDA to Begin Phase IIb Trial of Cancer Vaccine, Keytruda in Melanoma. Precision Medicine Online. Available online: <https://www.precisionmedicineonline.com/cancer/evaxion-cleared-fda-begin-phase-ii-b-trial-cancer-vaccine-keytruda-melanoma> (accessed on 23 October 2023).
168. Arnold, C. Inside the nascent industry of AI-designed drugs. *Nat. Med.* **2023**, *29*, 1292–1295. [Google Scholar] [CrossRef]
169. The Discovery of Amgen’s Novel Investigational KRAS(G12C) Inhibitor AMG 510 Published in Nature. Available online: <https://www.amgen.com/newsroom/press-releases/2019/10/the-discovery-of-amgens-novel-investigational-krasg12c-inhibitor-amg-510-published-in-nature> (accessed on 23 October 2023).
170. Ghislaine PELLAT. Constantin Anghelache. Governance in the EU Member States in the Era of Big Data. In Proceedings of the 25th PGV Network Conference—International Scientific Conference, Bucharest, Romania, 12–13 September 2019. Available online: https://www.researchgate.net/profile/Grzegorz-Maciejewski/publication/335929459_Use_of_Big_Data_On_The_Food_Market_-_Areas_Applications_Examples/links/5d84bb29a6fdcc8fd6fda856/Use-of-Big-Data-On-The-Food-Market-Areas-Applications-Examples.pdf (accessed on 2 January 2024).
171. Yang, X.; Wang, Y.; Byrne, R.; Schneider, G.; Yang, S. Concepts of Artificial Intelligence for Computer-Assisted Drug Discovery. *Chem. Rev.* **2019**, *119*, 10520–10594. [Google Scholar] [CrossRef] [PubMed]
172. How to Navigate the Patenting Challenges of AI-Assisted Drug Discovery. Available online: <https://www.pharmaceuticalonline.com/doc/how-to-navigate-the-patenting-challenges-of-ai-assisted-drug-discovery-0001> (accessed on 23 October 2023).
173. Freedman, D.H. Hunting for New Drugs with AI. *Nature* **2019**, *576*, S49–S53. [Google Scholar] [CrossRef] [PubMed]
174. How AI Is Aiming at the Bad Math of Drug Development. Bloomberg.com, 29 November 2021. Available online: <https://www.bloomberg.com/news/articles/2021-11-29/how-ai-is-aiming-at-the-bad-math-of-drug-development-quicktake> (accessed on 23 October 2023).
175. DiNuzzo, M. How artificial intelligence enables modeling and simulation of biological networks to accelerate drug discovery. *Front. Drug Discov.* **2022**, *2*, 1019706. Available online: <https://www.frontiersin.org/articles/10.3389/fddsv.2022.1019706> (accessed on 23 October 2023). [CrossRef]
176. Fleming, N. How artificial intelligence is changing drug discovery. *Nature* **2018**, *557*, S55–S57. [Google Scholar] [CrossRef]
177. Sahu A, Mishra J, Kushwaha N. Artificial Intelligence (AI) in Drugs and Pharmaceuticals. *Comb Chem High Throughput Screen.* **2022**;25(11):1818-1837. Doi: 10.2174/1386207325666211207153943. PMID: 34875986.

178. Singh S, Kumar R, Payra S, Singh SK. Artificial Intelligence and Machine Learning in Pharmacological Research: Bridging the Gap Between Data and Drug Discovery. *Cureus*. 2023 Aug 30;15(8):e44359. Doi: 10.7759/cureus.44359. PMID: 37779744; PMCID: PMC10539991.
179. Bekbolatova M, Mayer J, Ong CW, Toma M. Transformative Potential of AI in Healthcare: Definitions, Applications, and Navigating the Ethical Landscape and Public Perspectives. *Healthcare (Basel)*. 2024 Jan 5;12(2):125. Doi: 10.3390/healthcare12020125. PMID: 38255014; PMCID: PMC10815906.
180. Sarkar C, Das B, Rawat VS, Wahlang JB, Nongpiur A, Tiewsoh I, Lyngdoh NM, Das D, Bidarolli M, Sony HT. Artificial Intelligence and Machine Learning Technology Driven Modern Drug Discovery and Development. *Int J Mol Sci*. 2023 Jan 19;24(3):2026. Doi: 10.3390/ijms24032026. PMID: 36768346; PMCID: PMC9916967.
181. Gupta R, Srivastava D, Sahu M, Tiwari S, Ambasta RK, Kumar P. Artificial intelligence to deep learning: machine intelligence approach for drug discovery. *Mol Divers*. 2021 Aug;25(3):1315-1360. Doi: 10.1007/s11030-021-10217-3. Epub 2021 Apr 12. PMID: 33844136; PMCID: PMC8040371.
182. Yao T, Chen X, Wang H, Gao C, Chen J, Yi D, Wei Z, Yao N, Li Y, Yi D, Wu Y. Deep evolutionary fusion neural network: a new prediction standard for infectious disease incidence rates. *BMC Bioinformatics*. 2024 Jan 23;25(1):38. Doi: 10.1186/s12859-023-05621-5. PMID: 38262917; PMCID: PMC10804580.
183. Das K, Paltani M, Tripathi PK, Kumar R, Verma S, Kumar S, Jain CK. Current implications and challenges of artificial intelligence technologies in therapeutic intervention of colorectal cancer. *Explor Target Antitumor Ther*. 2023;4(6):1286-1300. Doi: 10.37349/etat.2023.00197. Epub 2023 Dec 27. PMID: 38213536; PMCID: PMC10776591.
184. Jinsong S, Qifeng J, Xing C, Hao Y, Wang L. Molecular fragmentation as a crucial step in the AI-based drug development pathway. *Commun Chem*. 2024 Feb 1;7(1):20. Doi: 10.1038/s42004-024-01109-2. PMID: 38302655; PMCID: PMC10834946.
185. Yan X, Gu C, Feng Y, Han J. Predicting Drug-drug Interaction with Graph Mutual Interaction Attention Mechanism. *Methods*. 2024 Jan 21;223:16-25. Doi: 10.1016/j.ymeth.2024.01.009. Epub ahead of print. PMID: 38262485.
186. Alkan C, O'Brien T, Kenyon V, Ikegami T. Computer-Selected Antiviral Compounds: Assessing In Vitro Efficacies against Rift Valley Fever Virus. *Viruses*. 2024 Jan 5;16(1):88. Doi: 10.3390/v16010088. PMID: 38257788; PMCID: PMC10818293.
187. Niazi SK, Mariam Z. Computer-Aided Drug Design and Drug Discovery: A Prospective Analysis. *Pharmaceuticals (Basel)*. 2023 Dec 22;17(1):22. Doi: 10.3390/ph17010022. PMID: 38256856; PMCID: PMC10819513.
188. Dawoud MHS, Mannaa IS, Abdel-Daim A, Sweed NM. Integrating Artificial Intelligence with Quality by Design in the Formulation of Lecithin/Chitosan Nanoparticles of a Poorly Water-Soluble Drug. *AAPS PharmSciTech*. 2023 Aug 8;24(6):169. Doi: 10.1208/s12249-023-02609-5. PMID: 37552427.
189. Vora LK, Gholap AD, Jetha K, Thakur RRS, Solanki HK, Chavda VP. Artificial Intelligence in Pharmaceutical Technology and Drug Delivery Design. *Pharmaceutics*. 2023 Jul 10;15(7):1916. Doi: 10.3390/pharmaceutics15071916. PMID: 37514102; PMCID: PMC10385763.
190. Raza MA, Aziz S, Noreen M, Saeed A, Anjum I, Ahmed M, Raza SM. Artificial Intelligence (AI) in Pharmacy: An Overview of Innovations. *Innov Pharm*. 2022 Dec 12;13(2):10.24926/iip.v13i2.4839. doi: 10.24926/iip.v13i2.4839. PMID: 36654703; PMCID: PMC9836757.
191. Douali L. Machine Learning to Predict Teratogenicity: Theory and Practice. *Methods Mol Biol*. 2024;2753:159-180. Doi: 10.1007/978-1-0716-3625-1_7. PMID: 38285338.

192. Swanson K, Walther P, Leitz J, Mukherjee S, Wu JC, Shivnaraine RV, Zou J. ADMET-AI: A machine learning ADMET platform for evaluation of large-scale chemical libraries. *bioRxiv* [Preprint]. 2023 Dec 28;2023.12.28.573531. doi: 10.1101/2023.12.28.573531. PMID: 38234753; PMCID: PMC10793392.
193. Parvatikar PP, Patil S, Khaparkhantkar K, Patil S, Singh PK, Sahana R, Kulkarni RV, Raghu AV. Artificial intelligence: Machine learning approach for screening large database and drug discovery. *Antiviral Res.* 2023 Dec;220:105740. Doi: 10.1016/j.antiviral.2023.105740. Epub 2023 Nov 5. PMID: 37935248.
194. Lv Q, Zhou F, Liu X, Zhi L. Artificial intelligence in small molecule drug discovery from 2018 to 2023: Does it really work? *Bioorg Chem.* 2023 Dec;141:106894. Doi: 10.1016/j.bioorg.2023.106894. Epub 2023 Sep 27. PMID: 37776682.

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